

Continuum Structural Complexity: Target-Aligned Source Algebra, Physical Expenditure, and Singular Reachability

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Abstract

A continuum structural presentation consists of a fixed PDE or flow, its legal orbits, a singular target in the state or germ space, the five channel families **Geom**, **Caus**, **Abs**, **Lift**, **Bdry**, and one finite physical expenditure budget. Each orbit carries a nonnegative countably additive measure μ_u that charges target-approaching events by physical action, dissipation, production, and admissible exterior work. The sole budget variable B_{phys} is a finite ceiling for its total mass. A common carrier identity derives this measure from energy, hypocoercive commutators, virial or positive-commutator laws, entropy production, and admissible work.

The physical-budget-saturated orbit algebra is obtained by completing raw orbit records sourcewise in the topology determined by the declared mathematical tests. Every weak limit, concentration profile, oscillation measure, defect measure, tail, boundary trace, and blow-up germ inherits the source word of the records that generate it. Normalization and chart changes append only **Abs** or **Lift**. Limit formation therefore creates no new structural channel. The residual coordinate is the common kernel of all prospective target functionals, so every target-visible dynamically generated record lies in an existing source cell.

Normalized target shells locate target progress before contact. If an orbit contacts the target, then for every sufficiently small shell it crosses a strictly pre-contact interval on which the shell observable rises exactly from zero to one. The chain rule and compiled current decomposition write this unit increment as the sum of five source integrals; the residual integral vanishes by target-nullity. Hence some inherited source current has positive shell pairing, in fact at least $1/5$. If no such singularity-aligned structure exists, the singular target is unreachable, irrespective of the size or compactness of the full orbit. Resolvent and stochastic identities provide parallel prospective source factorizations.

The remaining aligned branch is reduced sourcewise. Compatible null directions are quotiented once and propagated through all later certificates; each persistent source cell then needs only a target-closure invariant or a physical-price schedule with divergent total mass. Uniform positive gaps give a sufficient schedule. Hamiltonian resistance, passivity radii, capacity polarity, commutator carriers, constraint-tail cokernels, compactness-rigidity, scale renormalization, and Perelman-type entropy cocycles are compilers for these invariants. A finite null-or-paid source ledger excludes an infinite singular cascade. For a PDE whose target is its noncontinuation germ, the resulting theorem is a conditional global-regularity criterion that need not control target-null directions in a global norm.

Contents

Introduction	3
Visual architecture and proof-flow guide	4

Diagram map	4
Proof-dependency diagram	5
Node-by-node source audit	5
Monotone retained-fact table	7
Branch-closure audit	8
 I Closed Continuum Structural Algebra	 8
I.1 Physical continuum presentations	8
I.2 The five-source orbit algebra	11
I.3 Physical-budget-saturated sourcewise completion	13
I.4 Target quotient and the null residual	15
I.5 Prospective contact accountability	17
 II Physical Certificate Calculus	 23
II.1 Structural invariant reduction of analytic certificates	23
II.2 Covariant action and the physical budget	25
II.3 Hamiltonian meanings and chart covariance	28
II.4 Killed resolvents, resistance, and target realization	30
II.5 Full-generator response and physical aiming	33
II.6 Degenerate geometry and commutator-generated carriers	35
II.7 Compactness and rigidity as a price compiler	37
II.8 Constraint tails as finite structural obstructions	39
II.9 Covariant entropy cocycles and physical production	41
 III Structural Reachability and Regularity	 43
III.1 Singular targets and cascade compilers	43
III.2 The continuous structural regularity theorem	48
III.3 Certificate hierarchy and failure semantics	50
III.4 A reusable PDE instantiation protocol	52

III.5	Conclusion	53
A	Dependency table for the main conclusions	53
B	Notation index	55

Introduction

Let u be a legal orbit of a fixed continuum evolution and let Σ be a target in its state or germ space. The central question is whether the equation can reach Σ while spending only a finite amount of the physical quantity controlled by its evolution law. Each orbit carries one nonnegative countably additive expenditure measure μ_u on a declared event space $\mathfrak{Q}_u$, and the single budget assumption is

$$\mu_u(\mathfrak{Q}_u) \leq B_{\text{phys}} < \infty.$$

Action, dissipation, entropy production, forcing work, and boundary flux define this one measure; energy and conservation identities bound its total mass before the ceiling B_{phys} is set.

The structural source alphabet records how target-directed behavior is generated. The native dissipative or reversible mechanism is **Geom**; the fixed directed or nonreversible transport is **Caus**; quotient and symmetry reduction are **Abs**; charts, renormalizations, and augmented state variables are **Lift**; and boundary, terminal, killing, and exterior-flux records are **Bdry**. A composite continuum object may carry several labels, but every elementary source belongs to this alphabet.

The completion is sourcewise. A limit object is placed in the closure of the cell that generated it. Thus a concentration measure retains the ancestry of the concentrating quantity, a blow-up profile retains the ancestry of the physical orbit together with its **Abs/Lift** chart, and a terminal tail trace acquires **Bdry** ancestry. Failure to estimate a closed source cell leaves that estimate open; it does not create an additional continuum source.

The target-relative residual is correspondingly strict. Let $\mathcal{N}_\Sigma$ be the common kernel of the target shell derivatives, target fluxes, and other declared prospective contact functionals. The residual coordinate is contained in $\mathcal{N}_\Sigma$. Hence

$\text{target-visible} \implies \text{ancestry in } \{\text{Geom}, \text{Caus}, \text{Abs}, \text{Lift}, \text{Bdry}\}; \quad J_{\text{res}} \implies \text{target-null.}$
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Every target-visible limit therefore retains ancestry in one of the five source families.

The central target-relative question is therefore not whether the entire solution is large, but which source current contributes positively to the formation of the singularity. For a normalized shell crossing I_r , set

$$Q_{c,r} := \int_{I_r} \langle D\Phi_r(u_t), J_t^c \rangle dt.$$

Source completeness gives the exact pre-contact identity

$$1 = \sum_{c \in \mathfrak{C}} Q_{c,r}, \quad \sum_{c \in \mathfrak{C}} (Q_{c,r})_+ \geq 1, \quad \max_{c \in \mathfrak{C}} (Q_{c,r})_+ \geq \frac{1}{5}.$$

The residual source term vanishes. Thus contact itself supplies the unit progress and proves the existence of target-aligned structure. If all source currents vanish in the target-active quotient, their magnitude is irrelevant: every shell observable is constant along the orbit, so contact is excluded without compactness, a physical-price gap, or a global estimate.

The aligned branch is simplified before it is estimated. A certified target-null, gauge, or otherwise closed direction is passed to a compatible quotient and never reconsidered. What remains is summarized by four structural invariants: a target-closure space, a sharp domination radius, a physical-price gap, and a target-projected equality remainder. Many long analytic estimates are implementations of one of these checks after the source terms have been disentangled.

The main expenditure argument then uses the same measure μ_u . A singular contact compiler produces infinitely many pairwise disjoint pre-contact events and selects a persistent aligned source cell. Its ledger entry is either null, contradicting alignment, or physically paid, giving lower event prices whose sum diverges along an infinite subsequence. Countable additivity contradicts $\mu_u(\mathfrak{Q}_u) \leq B_{\text{phys}}$. The decisive application problem is to identify an expenditure measure whose lower prices have divergent total mass along every surviving target-approaching cascade. Basic energy alone need not have this property.

$\begin{aligned} &\text{contact} \xrightarrow{\text{normalized shell}} \text{target-aligned source,} \\ &\text{target-aligned source} \xrightarrow{\text{physical-price certificate}} \text{divergent expenditure,} \\ &\text{divergent expenditure} \implies \text{contradiction with } \mu_u(\mathfrak{Q}_u) \leq B_{\text{phys}}. \end{aligned}$

Hamiltonian, resolvent, commutator, constraint-tail, compactness, scale, and entropy arguments establish individual invariant rows. The symmetric form yields a spectral Hamiltonian and resistance; the full fixed-Caus generator requires nonnormal information; a carrier balance converts passivity into the single physical measure; compactness plus zero-price rigidity gives a gap; and a covariant entropy cocycle telescopes production across changing charts. These modules evaluate the five source families; they do not alter the source algebra.

Visual architecture and proof-flow guide

The twenty diagrams are maps of proved implications and certificate interfaces. Blue nodes contain evolution data, violet nodes contain ancestry or coordinate transformations, teal nodes contain geometric or compactness data, amber nodes contain target uses or decisions, coral nodes contain reachability outputs, and gray nodes contain target-null classes or certificate routes for which no estimate is assumed. Solid arrows denote implications or constructions. Dashed arrows denote comparisons or optional certificate routes.

The unique root of the global proof graph is a legal continuum presentation with a finite physical budget. Rectangles are live states with one successor, diamonds are exhaustive tests, and ellipses are terminal outcomes. Missing certificates terminate a proof method with an unevaluated source profile; they do not add a structural branch. All locality, source labels, and physical budget bounds established upstream remain active downstream. Node [6] begins the target-exclusion contradiction by assuming contact; the unit shell increment is then forced by the normalized shell definition.

Diagram map

No.	DAG node	Diagram	Principal internal sources
1	[1]–[13]	Global physical-reachability proof flow	Theorems I.3.4, I.4.5 and III.2.3
2	[1]	Physical orbit and target expenditure	Definitions I.1.1 and I.1.3
3	[1]	Physical expenditure budget	Definition I.1.3 and Proposition I.1.6
4	[2]	Five continuum source roles	Definition I.2.1, Theorem I.2.4, and Propositions I.2.5 and I.2.6
5	[3]	Sourcewise orbit completion	Definition I.3.2 and Theorem I.3.7
6	[4]	Limit ancestry and propagated reduction	Theorems I.3.4 and II.1.2
7	[5]	Target-visible re-entry and null residual	Definition I.4.2 and Theorem I.4.5
8	[6]–[7]	Normalized shell-crossing accountability	Theorems I.5.2 and I.5.12
9	[6]–[7]	Resolvent source-response factorization	Theorem I.5.5
10	[6]–[7]	Stochastic no-spontaneous-contact factorization	Theorems I.5.7 and I.5.12
11	[1], [9]–[10]	Carrier-to-physical-budget compiler	Theorems II.2.5, II.6.4 and II.6.6
12	[4]	Three Hamiltonian meanings	Proposition II.3.2
13	[4]	Same-current chart covariance	Theorem II.3.3 and Proposition II.3.4
14	[6], [9]	Killed arrival, resolvent, and resistance	Proposition II.4.3 and Theorems II.4.4 to II.4.6
15	[4], [6]	Full nonnormal target response	Definition II.5.1 and Proposition II.5.2
16	[6]–[9]	Physical aiming geometry	Theorem II.5.4 and Proposition II.5.5
17	[4], [9]	Estimate-to-invariant compiler	Theorems II.1.2, II.1.3, II.1.6 and II.7.4
18	[1], [9]–[10]	Covariant entropy-cocycle production	Definition II.9.1 and Theorem II.9.2
19	[6]–[11]	Target-aligned scale-cascade ledger	Proposition III.1.3, Theorems III.1.6 and III.1.13, and Corollary III.1.12
20	[1]–[11]	Structural global-regularity compiler	Theorem III.2.3 and Corollary III.2.5

Proof-dependency diagram

Node-by-node source audit

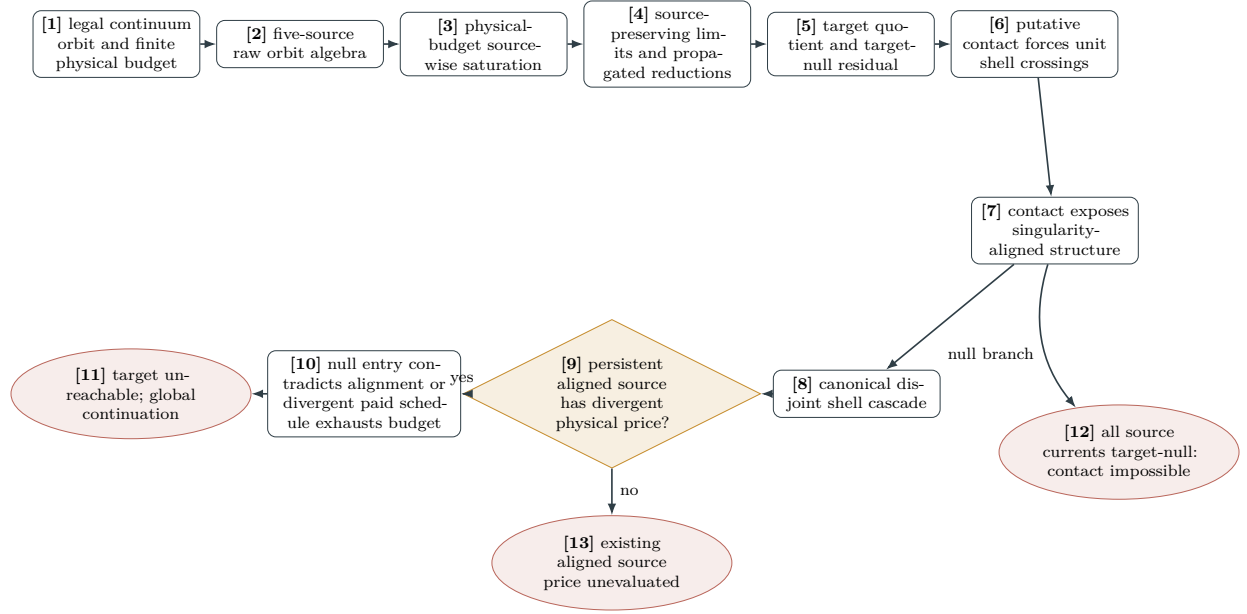


Figure 1: Global proof DAG. Nodes [1]–[5] construct the closed continuum algebra. At [6], putative contact forces exact unit shell crossings. At [7], contact itself exposes aligned structure; the target-null branch is therefore incompatible with contact. Node [8] extracts disjoint shell events without a compactness hypothesis. Nodes [9]–[10] evaluate only the physical price of the persistent aligned source. Failure of that evaluation leaves an existing source profile open and does not alter the taxonomy.

Node	State or test	Exact source	Successor
[1]	Physical presentation, carrier, and finite expenditure measure	Definitions I.1.1 and I.1.3 and Theorem II.2.5	[2]
[2]	Raw five-source orbit algebra	Definition I.2.1	[3]
[3]	Sourcewise physical-budget saturation	Definition I.3.2	[4]
[4]	Limit ancestry and certified reductions propagate	Theorems I.3.4 and II.1.2	[5]
[5]	Target-active quotient and null residual	Definition I.4.2 and Theorem I.4.5	[6]
[6]	Putative contact and unit pre-contact shell crossings	Definition I.5.1 and Theorems I.5.2 and I.5.10	[7]
[7]	Singularity-formation support test	Theorem I.5.12 and Proposition III.1.3	[8] or [11]
[8]	Canonical disjoint shell cascade	Theorem III.1.4 and Definition III.1.5	[9]
[9]	Persistent-source physical-price test	Theorem II.1.6 and Eq. (127)	[10] or [13]
[10]	Persistent source is null or has a divergent paid schedule	Theorem III.1.13	[11]
[11]	Target exclusion and PDE continuation	Theorem III.2.3 and Corollary III.2.5	terminal
[12]	Every structural current is target-null	Corollary I.5.3	terminal
[13]	Persistent-source price not established	Remark III.3.1	terminal

Monotone retained-fact table

Retained fact	Persistence rule
Physical orbit	Every normalization, window, limit, and prospective functional is attached to the same legal evolution.
Physical budget	Restrictions to subintervals and pairwise disjoint event families use one nonnegative expenditure measure and cannot increase its total mass.
Source ancestry	Algebraic operations take unions of source words; sourcewise limits remain in the same closed cell.
Certified reduction	A closed target-null or otherwise excluded direction is propagated through every compatible quotient and is never re-estimated downstream.
Fixed Caus	Charts transport the declared nonreversible current and do not replace it by a target probe or an optimizer.

Target residual	The residual remains in the common target kernel under closure; a nonzero target pairing triggers re-entry into a source cell.
Target alignment	Every normalized contact shell has unit total progress, so at least one source pairing is at least 1/5; target-null motion may be arbitrarily large without entering later estimates.
Source evaluation	Each persistent source cell is reduced exactly once, by a target-null invariant or by a physical-price invariant.
Locality	Compactness, chain rules, lower semicontinuity, and price bounds are used only on the windows and topologies named by their certificates.
Certificate status	Failure to bound a cell changes its bound status only; its source label remains fixed.

Branch-closure audit

Route	Branch	Closure source	Output
[7]–[11]	no singularity-aligned source occurs	Theorem I.5.12 and Proposition III.1.3	target unreachable directly
[7]–[12]	every compiled source pairing is target-null	Corollary I.5.3	target unreachable directly
[9]–[13]	the persistent source has no proved divergent price	Remark III.3.1	existing source price unevaluated
[9]–[10]–[11]	every persistent source has a null entry or divergent paid schedule	Theorems III.1.13 and III.2.3	target unreachable

Part I

Closed Continuum Structural Algebra

I.1 Physical continuum presentations

Definition I.1.1 (Legal physical continuum presentation). A legal physical continuum presentation is a tuple

$$\mathfrak{P}_{\text{phys}} = (X, \mathfrak{U}, \mathfrak{G}, \mathfrak{C}, B_{\text{phys}}, \mu, \Sigma, \mathfrak{F}_{\Sigma})$$

with the following data.

- (P1) X is a Hausdorff state space, possibly accompanied by stronger regularity spaces and a germ space X_{germ} .
- (P2) $\mathfrak{U}$ is a class of maximal legal orbits $u : [0, T_*(u)) \rightarrow X$ determined by one fixed evolution law and solution concept. The presentation specifies restriction, concatenation when uniqueness permits it, and the continuation criterion used to define $T_*(u)$.
- (P3) $\mathfrak{G}$ is the declared symmetry and gauge family. Its action on states, currents, physical expenditure, and target germs is specified before quotienting.

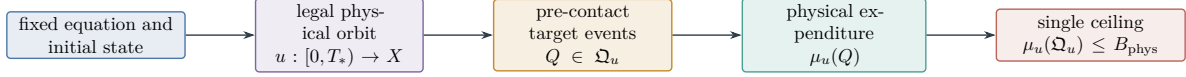


Figure 2: One physical-budget architecture. The equation generates the orbit, target approach selects pre-contact events, and all such events spend the same countably additive measure whose total mass is bounded by B_{phys} .

(P4)

$$\mathfrak{C} = \{\text{Geom}, \text{Caus}, \text{Abs}, \text{Lift}, \text{Bdry}\}$$

is the inherited source alphabet of Definition I.2.1. Native terms are routed by the sector/gradient compiler and subsequent orbit-record operations acquire their source words by the constructor grammar; this is proved in Theorem I.4.4.

- (P5) $B_{\text{phys}} < \infty$ is the physical expenditure ceiling. Every orbit under that ceiling carries one nonnegative countably additive measure μ_u satisfying Definition I.1.3. Its measurable sets may be time intervals, spacetime cylinders, scale–time windows, or another declared event family. Every nonnegative physical quantity used in the target exclusion argument is included in this measure.
- (P6) $\Sigma \subset X \cup X_{\text{germ}}$ is the target. The family $\mathfrak{F}_\Sigma$ contains the continuous target-shell, flux, or contact functionals used to detect approach to Σ , together with the differentiability or generator-domain data needed by the prospective formulas.

The presentation is fixed by the equation, its legal solution class, the target, and the physical expenditure law.

Remark I.1.2 (Fixed dynamics). The term “legal” refers to the solution class and the declared physical boundary or forcing data. It does not introduce a policy that selects among arbitrary state transitions. For an autonomous PDE, the equation fixes the vector field.

Definition I.1.3 (Physical expenditure budget). Let $u \in \mathfrak{U}$. Its physical expenditure is a nonnegative measure μ_u on a measurable event space $(\Omega_u, \mathcal{A}_u)$. The total mass $\mu_u(\Omega_u)$ is the orbit’s actual physical expenditure. The sole budget ceiling $B_{\text{phys}} < \infty$ admits precisely the orbits satisfying

$$\mu_u(\Omega_u) \leq B_{\text{phys}} < \infty. \quad (1)$$

The measure is *localizable*: the price of a disjoint family is the sum of its prices. Thus, for pairwise disjoint $Q_j \in \mathcal{A}_u$,

$$\sum_j \mu_u(Q_j) = \mu_u\left(\bigsqcup_j Q_j\right) \leq B_{\text{phys}}. \quad (2)$$

If the evolution supplies several nonnegative expenditures, the presentation fixes their units and coefficients and combines them into μ_u before B_{phys} is declared. Every event price used in an exclusion argument is therefore charged to this one measure.

Example I.1.4 (Energy–work budget). Suppose an energy $\mathcal{E}$ is bounded below by E_- , a nonnegative dissipation measure μ_{diss} satisfies

$$\mathcal{E}(u(t)) + \mu_{\text{diss}}([0, t]) \leq \mathcal{E}(u_0) + W^+([0, t]),$$

and $W^+([0, \infty)) < \infty$. Then

$$\mu_{\text{diss}}([0, \infty)) \leq \mathcal{E}(u_0) - E_- + W^+([0, \infty)).$$

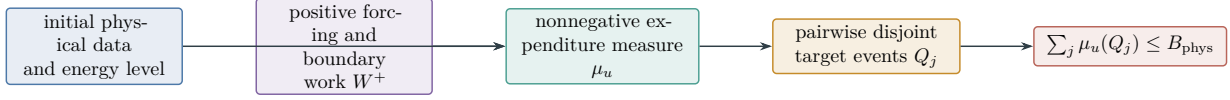


Figure 3: Physical expenditure accounting. The initial state and legitimate positive input bound one countably additive measure. Every disjoint target event spends that same measure, whose total mass is the sole budget.

Taking $\mu_u = \mu_{\text{diss}}$ gives the single budget

$$B_{\text{phys}} = \mathcal{E}(u_0) - E_- + W^+([0, \infty)).$$

A stronger scale-sensitive expenditure measure may be required when finite basic energy does not charge singular concentration.

Definition I.1.5 (Fixed-Caus gradient-flow presentation). Let $\mathcal{V} \hookrightarrow \mathcal{H} \hookrightarrow \mathcal{V}^*$ be a Gelfand triple. A fixed-Caus gradient-flow presentation consists of a differentiable energy $\mathcal{E}$, a symmetric nonnegative mobility $\mathbb{K}(u) : \mathcal{V}^* \rightarrow \mathcal{V}$, a fixed constitutive transport $A_{\text{Caus}}(u) \in \mathcal{V}^*$, and an exterior input $f_{\text{Bdry}} \in \mathcal{V}^*$, with equation

$$\dot{u} + A_{\text{Caus}}(u) = -\mathbb{K}(u)D\mathcal{E}(u) + f_{\text{Bdry}}. \quad (3)$$

The Caus term is energy-skew when

$$\langle D\mathcal{E}(u), A_{\text{Caus}}(u) \rangle = 0. \quad (4)$$

Write

$$\|\mathbb{K}(u)^{1/2}\xi\|^2 := \langle \xi, \mathbb{K}(u)\xi \rangle$$

for $\xi \in \mathcal{V}^*$; this notation does not assume a bounded inverse or a globally defined square root on $\mathcal{V}^*$. Here **Geom** is the mobility current $-\mathbb{K}D\mathcal{E}$, **Caus** is the declared transport, and **Bdry** contains the exterior input and boundary flux. Quotients and symmetry reductions applied to this orbit have **Abs** ancestry; charts, blow-up rescalings, and augmented variables have **Lift** ancestry.

Proposition I.1.6 (Fixed-Caus expenditure bound). *Let u satisfy Eqs. (3) and (4) on $[0, T]$, with the chain rule for $\mathcal{E}(u)$. Then*

$$\frac{d}{dt}\mathcal{E}(u(t)) = -\|\mathbb{K}(u)^{1/2}D\mathcal{E}(u)\|^2 + \langle D\mathcal{E}(u), f_{\text{Bdry}} \rangle. \quad (5)$$

If $\mathcal{E} \geq E_-$, then

$$\int_0^T \|\mathbb{K}(u)^{1/2}D\mathcal{E}(u)\|^2 dt \leq \mathcal{E}(u_0) - E_- + \int_0^T \langle D\mathcal{E}(u), f_{\text{Bdry}} \rangle_+ dt. \quad (6)$$

For a weak solution with an energy inequality, the same statement holds with an additional nonnegative energy-defect measure on the left.

Proof. Pair (3) with $D\mathcal{E}(u)$. The chain rule gives the left side of (5), the Caus pairing vanishes by (4), and positivity of $\mathbb{K}$ gives the squared norm. Integrate, discard the terminal energy above E_- , and replace exterior work by its positive part. The weak form is the increment formula for the energy inequality; its defect is nonnegative. $\square$

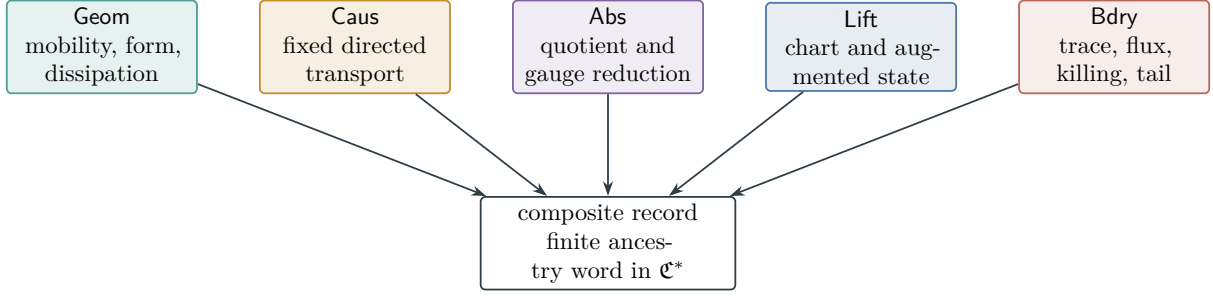


Figure 4: The five continuum source roles. Geom and Caus are native current mechanisms; Abs and Lift record transformations of those mechanisms; Bdry records exterior and terminal effects. Composite records retain the union of their ancestries.

I.2 The five-source orbit algebra

Definition I.2.1 (Continuum source alphabet). The source alphabet is

$$\mathfrak{C} = \{\text{Geom}, \text{Caus}, \text{Abs}, \text{Lift}, \text{Bdry}\}.$$

A raw continuum record is an orbit occurrence together with a finite ancestry word $\text{Anc}(r) \in \mathfrak{C}^*$. Its source support $\text{Src}(r) \subseteq \mathfrak{C}$ is the set of letters appearing in that word. The primitive and closure rules are:

- (S1) symmetric mobility, Dirichlet-form, reversible, and dissipative current records start in **Geom**;
- (S2) the fixed directed transport and its current start in **Caus**;
- (S3) quotient, symmetry reduction, gauge removal, and coarse variables append **Abs**;
- (S4) charts, renormalizations, blow-up variables, auxiliary-state extensions, and Young- or defect-variable lifts append **Lift**;
- (S5) boundary traces, exterior fluxes, killing, terminal values, and tails read at the boundary of a compactification append **Bdry**;
- (S6) finite algebraic combination, restriction, and concatenation take the union of the input source supports.

The symbol J_{res} is absent from this source alphabet. It is introduced only as the target-null coordinate of Definition I.4.2.

Remark I.2.2 (Channel and source terminology). The channels **Geom** and **Abs** carry intrinsic reusable structure, while **Caus**, **Lift**, and **Bdry** orient, reorganize, and read out that structure. A *source cell* is a provenance cell in the complete five-channel normal form; this terminology does not promote the three derived channels to independent creators of intrinsic structure.

Remark I.2.3 (Forcing and constraints). The presentation classifies a volume force according to its constitutive role. A fixed directed body force may be **Caus**, while exterior work or a boundary supply is **Bdry**. A Lagrange multiplier implementing a quotient constraint has **Abs** ancestry. The five labels are assigned from the declared weak equation, not from the sign of a later estimate.

Theorem I.2.4 (Sector-form source compiler). *Let X be locally compact and separable, let m be a Radon measure of full support, and let $(\mathcal{E}^s, D(\mathcal{E}))$ be a regular symmetric Dirichlet form on the real space $L^2(X, m)$. Write its Beurling–Deny decomposition, with normalization absorbed into J , as*

$$\mathcal{E}^s(u, v) = \mathcal{E}^c(u, v) + \int_{X \times X \setminus \text{diag}} (u(x) - u(y))(v(x) - v(y)) J(\text{d}x, \text{d}y) + \int_X uv \kappa(\text{d}x). \quad (7)$$

Let $\mathcal{E}^a$ be antisymmetric on the same domain and satisfy the sector bound

$$|\mathcal{E}^a(u, v)| \leq C_{\text{sec}} \mathcal{E}_1^s(u, u)^{1/2} \mathcal{E}_1^s(v, v)^{1/2}. \quad (8)$$

Then the form $\mathcal{E} = \mathcal{E}^s + \mathcal{E}^a$ is closed and sectorial on $D(\mathcal{E})$, and its source assignment is exhaustive: $\mathcal{E}^c$ and the symmetric jump form are **Geom**, $\mathcal{E}^a$ is the fixed **Caus** form, and the killing measure κ is **Bdry**. The uniqueness of (7) leaves no additional symmetric Markovian primitive. Quotients and state-space enlargements append **Abs** and **Lift** as usual.

Proof. The Beurling–Deny theorem gives (7) and uniqueness of its strongly local, jump, and killing measures [3]. Antisymmetry gives $\mathcal{E}^a(u, u) = 0$, while (8) makes it continuous in the $\mathcal{E}_1^s$ -norm. Hence $\mathcal{E}$ has the same form norm completeness as $\mathcal{E}^s$ and satisfies the sector condition. The stated routing is exactly the source assignment of Definition I.2.1. A quasi-regular form has the same conclusion after a declared quasi-homeomorphic regular realization. $\square$

Proposition I.2.5 (Exact-drift symmetrization). *Let M be a Riemannian domain, let $\Psi \in C^2(M)$, and suppose $\pi(dx) = e^{-\Psi(x)} dx$ is a locally finite measure. For*

$$L_\Psi g = \Delta g - \nabla \Psi \cdot \nabla g,$$

assume compact support or boundary conditions that eliminate the integration by parts trace. Then, on the resulting form core,

$$-\langle f, L_\Psi g \rangle_{L^2(\pi)} = \int_M \nabla f \cdot \nabla g \, d\pi. \quad (9)$$

*Thus an exact drift is absorbed into a symmetric **Geom** form by a weighted **Abs**-gauge. If an exact/nonexact decomposition of a general drift is proved on the declared domain, only its nonexact remainder retains independent **Caus** ancestry. No global boundedness of $e^{\pm\Psi}$ is inferred; any comparison with another physical measure must be proved separately.*

Proof. The weighted divergence identity

$$L_\Psi g = e^\Psi \operatorname{div}(e^{-\Psi} \nabla g)$$

and the stated trace condition give (9). Symmetry and nonnegativity of the right side prove the source statement; changing the reference measure is the declared gauge operation. $\square$

Proposition I.2.6 (Wiener lift creates no additional source). *Let H and U be separable Hilbert spaces and let a legal Itô evolution have drift $b : H \rightarrow H$ and Hilbert–Schmidt noise coefficient $\sigma : H \rightarrow \mathcal{L}_2(U, H)$. On bounded C^2 cylinder functions, assume its generator is*

$$LF(u) = \langle DF(u), b(u) \rangle + \frac{1}{2} \operatorname{Tr}[\sigma(u)\sigma(u)^* D^2 F(u)]. \quad (10)$$

Then the carré du champ is

$$\Gamma(F) := L(F^2) - 2FLF = \|\sigma^* DF\|_U^2 \geq 0. \quad (11)$$

*Thus the diffusion covariance and predictable quadratic variation are **Geom**; the drift retains its declared **Geom/Caus/Bdry** decomposition; and adjoining the Wiener path is **Lift**. Noise therefore creates no sixth continuum source.*

Proof. Apply the second-order chain rule to F^2 in (10); the drift terms cancel and the remaining trace is $\|\sigma^* DF\|_U^2$. Itô’s formula gives the same integrand for the predictable quadratic variation of the martingale part. The source labels follow from positivity of Γ , the fixed drift decomposition, and the state-space augmentation rule in Definition I.2.1. $\square$

I.3 Physical-budget-saturated sourcewise completion

The closed continuum algebra must contain every mathematical limit generated by a legal finite-budget orbit. Compactness in a strong PDE topology is a separate certificate question. The following evaluation completion supplies the closed algebra without asserting strong convergence of nonlinear terms.

Definition I.3.1 (Structural test system). For each nonempty $A \subseteq \mathfrak{C}$ and physical ceiling B_{phys} , let $\mathcal{R}_A^0(B_{\text{phys}})$ be the raw orbit records with source support equal to A . A structural test system is a family $\mathfrak{F}_A$ of maps

$$\phi : \mathcal{R}_A^0(B_{\text{phys}}) \longrightarrow K_\phi,$$

where every K_ϕ is compact Hausdorff. The family separates the records at the resolution used by the theory and contains:

- (T1) the weak PDE pairings and source-current pairings;
- (T2) the target shells and target fluxes;
- (T3) the physical-price and entropy-production readouts used downstream;
- (T4) the chart, quotient, boundary, and tail coordinates whose persistence is required by a later theorem.
- (T5) the graphs of every finite algebraic operation used downstream; each such operation is required to extend continuously between the named product cells.

An unbounded real-valued test is inserted through a separating family of bounded continuous truncations or through a fixed injective compactification of $\mathbb{R}$. Thus every coordinate target remains compact.

Definition I.3.2 (Physical-budget-saturated closed orbit algebra). Let

$$\iota_A(r) = (\phi(r))_{\phi \in \mathfrak{F}_A} \in K_A := \prod_{\phi \in \mathfrak{F}_A} K_\phi.$$

The space K_A is compact Hausdorff by Tychonoff's theorem. The sourcewise saturated cell and the full physical orbit algebra are

$$\mathcal{R}_A^{\text{phys}, \text{sat}}(B_{\text{phys}}) := \overline{\iota_A(\mathcal{R}_A^0(B_{\text{phys}}))}^{K_A}, \quad \mathcal{R}^{\text{phys}, \text{sat}}(B_{\text{phys}}) := \bigsqcup_{\emptyset \neq A \subseteq \mathfrak{C}} \mathcal{R}_A^{\text{phys}, \text{sat}}(B_{\text{phys}}). \quad (12)$$

A point of the closure is a mathematical limit record. Its ancestry support is the cell label A . A finer ancestry word may also be retained when it is stable under the chosen test system. The closure contains all limits of legal orbit records seen by the declared tests. Here Src denotes provenance support, not the smallest support of an analytically simplified formula.

Remark I.3.3 (Why the disjoint union is retained). The same analytic denotation can arise through different source histories. The closed algebra retains those histories as distinct records. Forgetting ancestry is an evaluation map from the disjoint union to the ambient analytic space; it is not part of source classification.

Theorem I.3.4 (Source-preserving continuum completion). *Let (r_i) be a net of legal orbit records with $\text{Src}(r_i) = A$, and suppose it converges in the structural test topology. Its limit record belongs to $\mathcal{R}_A^{\text{phys}, \text{sat}}(B_{\text{phys}})$. Consequently*

$$\text{Src}(\lim_i r_i) = A = \bigcup_i \text{Src}(r_i). \quad (13)$$

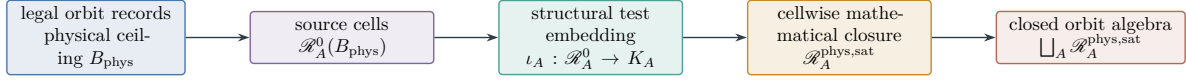


Figure 5: Sourcewise physical saturation. Closure is taken separately in each ancestry cell before the cells are assembled, and every cell retains the same physical ceiling B_{phys} .

If the net first passes through quotient or chart maps, **Abs** or **Lift**, respectively, is appended before taking the limit. If the limit is read as a boundary or tail trace, **Bdry** is appended. Thus defect measures, oscillation variables, concentration profiles, closure directions, blow-up germs, and tails create no source outside $\mathfrak{C}$.

Proof. By definition, $\mathcal{R}_A^{\text{phys,sat}}(B_{\text{phys}})$ is the closure of the test image of the raw cell with provenance support A . Hence every limit of such a net belongs to that cell. The algebraic rules of Definition I.2.1 take unions of source supports and append the labels of quotient, chart, and boundary operations before the closure is formed. The new coordinates of a limit record are limits of tests on the same tagged net; they do not supply a new primitive letter. This proves (13) and each listed case. $\square$

Corollary I.3.5 (Mixed sequences). *Every net of raw records has source support in the finite set $2^{\mathfrak{C}} \setminus \{\emptyset\}$. Any convergent subnet can be refined to a subnet with constant source support. Each resulting limit is therefore contained in an existing saturated cell.*

Proof. There are only $2^5 - 1$ nonempty source supports. One support is cofinal along a subnet. Apply Theorem I.3.4 to that subnet. $\square$

Definition I.3.6 (Source realization map). Let $\mathcal{P}_A$ and $\mathcal{J}_A$ be Hilbert spaces of potentials and currents for a fixed source support $A \subseteq \mathfrak{C}$. A *source realization map* is a densely defined closed operator

$$G_A : D(G_A) \subseteq \mathcal{P}_A \longrightarrow \mathcal{J}_A$$

whose range consists of the currents carrying an actual potential witness in that source cell. Its saturated current space is $\overline{\text{Ran } G_A}^{\mathcal{J}_A}$. The closure has the same source support A , independently of whether every closure direction has a potential witness.

Theorem I.3.7 (Closed-range realization criterion). *Let $G : D(G) \subseteq \mathcal{P} \rightarrow \mathcal{J}$ be a source realization map. The following conditions are equivalent.*

- (R1) $\text{Ran } G$ is closed in $\mathcal{J}$.
- (R2) There is $\gamma > 0$ such that

$$\|G\phi\|_{\mathcal{J}}^2 \geq \gamma \text{dist}_{\mathcal{P}}(\phi, \ker G)^2, \quad \phi \in D(G). \quad (14)$$

- (R3) The restriction of G^*G to $(\ker G)^\perp$ has spectrum in $[\gamma, \infty)$ for some $\gamma > 0$.
- (R4) The Moore–Penrose inverse G^+ is bounded on $\text{Ran } G$.

When these conditions hold, every saturated current in $\overline{\text{Ran } G}$ has a potential witness, unique modulo $\ker G$. When they fail, elements of $\overline{\text{Ran } G} \setminus \text{Ran } G$ remain limit records in the same source cell by Theorem I.3.4; failure of potential realization does not define another source family.

Proof. The closed-range theorem for densely defined closed operators gives the equivalence of (R1), (R2), and boundedness of G^+ . On $(\ker G)^\perp$, $\|G\phi\|^2 = \langle G^*G\phi, \phi \rangle$, so the spectral theorem makes (R2) equivalent to (R3). Closed range gives $\overline{\text{Ran } G} = \text{Ran } G$, and the kernel is exactly the ambiguity in a potential witness. The final assertion is Theorem I.3.4 applied to the source cell generated by G . $\square$

Theorem I.3.8 (Target-relative gradient exhaustiveness). *Let $G : D(G) \subseteq \mathcal{P} \rightarrow \mathcal{J}$ be a densely defined closed realization of one inherited source cell, and let $P_{\overline{G}}$ be the orthogonal projection onto $\overline{\text{Ran } G}$. Then every $j \in \mathcal{J}$ has the unique split*

$$j = P_{\overline{G}}j + (I - P_{\overline{G}})j, \quad P_{\overline{G}}j \in \overline{\text{Ran } G}, \quad (I - P_{\overline{G}})j \in \ker G^*. \quad (15)$$

The first term belongs to the saturated cell generated by G , including when it lies in $\overline{\text{Ran } G} \setminus \text{Ran } G$. Relative to the target family Γ_Σ , the second term has the further canonical split into its class in $\ker G^/(\ker G^* \cap \mathcal{N}_\Sigma)$ and its target-null part. Hence failure of closed range creates neither a new source nor a target-visible residual: it creates a closure direction in the inherited cell, while every orthogonal target-visible class remains a current of the native compiled equation and retains its native ancestry.*

Proof. The range closure is a closed Hilbert subspace, so the projection theorem gives (15); the identity $(\text{Ran } G)^\perp = \ker G^*$ identifies its complement. The definition of the sourcewise completion and Theorem I.3.4 assign $P_{\overline{G}}j$ to the same tagged cell as the approximating gradients. Quotienting the complementary space by its intersection with $\mathcal{N}_\Sigma$ is exact. Its kernel is target-null by definition, and a nonzero quotient class has a nonzero target pairing. Such a class cannot be renamed residual; its ancestry is the ancestry of the native current from which j was compiled by Theorem I.4.4. These alternatives exhaust $\mathcal{J}$. $\square$

Corollary I.3.9 (Regularized realization). *For $j \in \mathcal{J}$ and $\lambda > 0$, the functional*

$$\phi \mapsto \|G\phi - j\|_{\mathcal{J}}^2 + \lambda \|\phi\|_{\mathcal{P}}^2$$

has a unique minimizer $\phi_\lambda \in D(G)$. The resulting current $G\phi_\lambda$ is a stable regularized realization, with the parameter λ recording its bias. Existence of this minimizer for every $\lambda > 0$ does not imply any of the equivalent conditions in Theorem I.3.7.

Proof. On $D(G)$, the form $\langle G\phi, G\psi \rangle + \lambda \langle \phi, \psi \rangle$ is closed and coercive in its form norm. The functional $\psi \mapsto \langle j, G\psi \rangle$ is continuous in that norm, so the Riesz representation theorem gives a unique minimizer. The final statement follows because adding λI supplies coercivity even when the unregularized range is not closed. $\square$

Remark I.3.10 (Structural and certificate topologies). The evaluation compactification closes the structural algebra. A nonlinear PDE estimate may require a stronger topology to pass a particular term to the limit or to keep a bad set closed. Establishing that stronger compactness is a certificate task in Section II.7; its failure does not remove the limit record from its source cell.

I.4 Target quotient and the null residual

Definition I.4.1 (Prospective target functionals). Let $\mathcal{J}^{\text{phys, sat}}(B_{\text{phys}})$ be the current-occurrence part of the closed orbit algebra. Assume its finite-current locus is a real Hausdorff locally convex space under the continuously extended addition and scalar multiplication. Records at an infinite compactification endpoint are treated as boundary or tail records and are tested by their bounded readouts, not inserted into this vector space. A prospective target family Γ_Σ is a nonempty normalized collection of continuous real-linear functionals on this current space. It contains every pairing used to measure pre-contact progress, including

$$J \mapsto \langle D\Phi_r(z), J \rangle$$

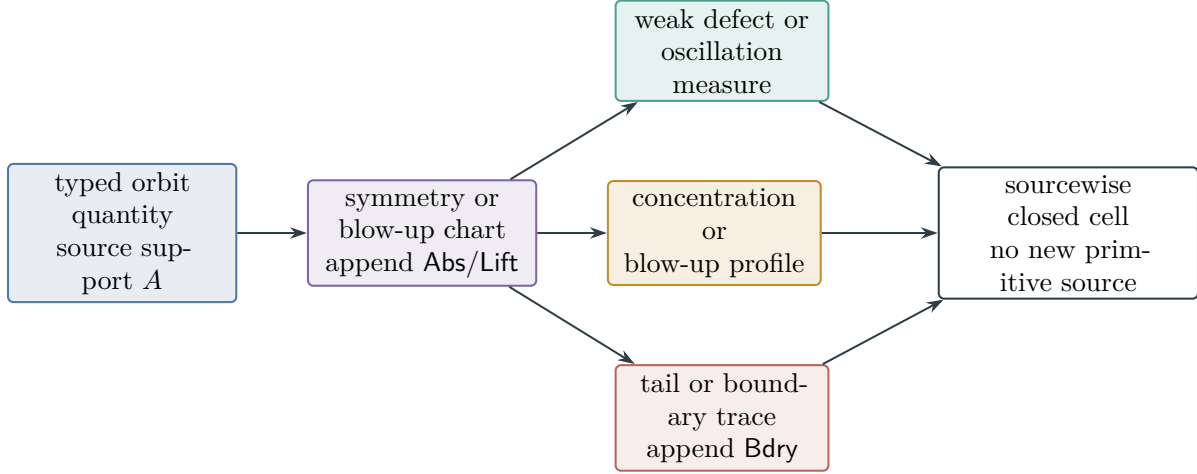


Figure 6: Continuum limit phenomena inherit ancestry. Normalization changes the source word in the declared Abs/Lift manner; a terminal trace appends Bdry. Weakness or concentration changes the analytic realization, not the source alphabet.

for the declared differentiable target shells, and the analogous killed-flux or compensator pairings when those interfaces are used. The normalization is part of the target geometry; for example, one may require unit dual norm.

Definition I.4.2 (Target-null residual). The target-null space and target-active quotient are

$$\mathcal{N}_\Sigma := \bigcap_{\Lambda \in \Gamma_\Sigma} \ker \Lambda, \quad \mathcal{J}_\Sigma := \mathcal{J}^{\text{phys,sat}}(B_{\text{phys}})/\mathcal{N}_\Sigma. \quad (16)$$

The post-saturation residual is any closed current coordinate containing zero and satisfying

$$\mathcal{J}_{J_{\text{res}}}^{\text{phys,sat}} \subseteq \mathcal{N}_\Sigma. \quad (17)$$

It is an audit coordinate for target-invisible transport, not a sixth source family. A current is target-visible precisely when its quotient class in $\mathcal{J}_\Sigma$ is nonzero.

Definition I.4.3 (Compiled continuum current). A continuum current is *compiled* when it is generated from the native terms of the fixed equation by the finite constructor rules (S1)–(S6) of Definition I.2.1. Its evaluation has the normal form

$$J = J^{\text{Geom}} + J^{\text{Caus}} + J^{\text{Abs}} + J^{\text{Lift}} + J^{\text{Bdry}} + J^{J_{\text{res}}}, \quad J^{J_{\text{res}}} \in \mathcal{N}_\Sigma, \quad (18)$$

as an equality of weak current pairings. Empty components are allowed. The residual is an optional regrouping in the common target kernel after the five source components have been evaluated; the choice $J^{J_{\text{res}}} = 0$ is always admissible. It is not a constructor and carries no source letter.

Theorem I.4.4 (Continuum normal-form compilation). *Suppose the native weak evolution is either a fixed-Caus gradient-flow presentation in the sense of Definition I.1.5 or is generated by a sector form satisfying Theorem I.2.4. Adjoin only the declared quotient, chart, auxiliary-state, trace, forcing, and boundary operations from Definition I.2.1. Then every finite orbit-current expression is compiled and has (18). After sourcewise completion, every finite target pairing has the same normal form. Thus the five-channel exhaustion is a theorem of the inherited constructor algebra.*

Proof. In the fixed-Caus case, positivity of $\mathbb{K}$, the declared directed transport, and the exterior term give **Geom**, **Caus**, and **Bdry**, respectively. In the form case, the sector-form compiler makes the same assignment to the symmetric mobility, antisymmetric directed, and killing terms. Structural induction on the syntax of an orbit-current expression now proves the claim. A quotient appends **Abs**, a chart or auxiliary variable appends **Lift**, a trace appends **Bdry**, and finite combination takes the union of the already assigned supports. These are all constructors of the algebra, so the induction has no further case.

For a sourcewise convergent net, each prospective target functional is a coordinate of the structural test system. Its value therefore passes to the limit. By Theorem I.3.4, the limit remains in the same tagged source cell. The image in the quotient by $\mathcal{N}_\Sigma$ is therefore generated by the five cell images. Taking $J^{\text{res}} = 0$, or regrouping any already identified target-null summand as J^{res} , gives (18) without requiring a complemented kernel and without adding a primitive source. $\square$

Theorem I.4.5 (No post-saturation target-visible residual). *Let $\mathfrak{P}_{\text{phys}}$ be generated by the compiled continuum algebra of Theorem I.4.4. Every target-visible current record in $\mathcal{J}^{\text{phys,sat}}(B_{\text{phys}})$ has nonempty ancestry in $\mathfrak{C}$, and its target-active quotient is generated by the images of the five saturated source cells. The residual source profile vanishes:*

$$G_{B_{\text{phys}},\Sigma}^{J^{\text{res,src,sat}}} := \sup_{J \in \mathcal{J}_{\text{res}}^{\text{phys,sat}}} \sup_{\Lambda \in \Gamma_\Sigma} (\Lambda J)_+ = 0. \quad (19)$$

If a provisional limit object pairs nontrivially with a target functional, then Theorem I.3.4 places it in the existing source cell inherited from its generating records. Thus every dynamically generated target-visible limit belongs to one of the five source families.

Proof. For raw occurrences, apply the quotient map $q_\Sigma : \mathcal{J}^{\text{phys,sat}} \rightarrow \mathcal{J}_\Sigma$ to (18). The residual term vanishes because it lies in $\mathcal{N}_\Sigma$. Continuity of every $\Lambda \in \Gamma_\Sigma$ passes the same identity to each sourcewise limit. The source-preserving theorem assigns every nonresidual limit term to the closure of its existing cell. Thus a nonzero quotient class is generated by the five cell images. Finally, every functional in Γ_Σ vanishes on $\mathcal{J}_{\text{res}}^{\text{phys,sat}}$, so both nested suprema in (19) are zero. $\square$

Corollary I.4.6 (Source type and bound status are independent). *Each saturated source cell may be marked bounded, unbounded, or unevaluated by a chosen certificate family. This mark changes no source word and no target quotient. In particular, failure to prove compactness or a resolvent estimate leaves the corresponding $G_{B_{\text{phys}},\Sigma}^{A,\text{src,sat}}$ unevaluated; it does not move the cell into J_{res} .*

I.5 Prospective contact accountability

Target contact may produce a retrospective object, such as a singular germ or a completed hitting distribution. Source accountability is instead read from the evolution strictly before contact. The next three theorems give deterministic, semigroup, and stochastic versions of the same factorization.

Definition I.5.1 (Target-shell presentation). Let $\hat{X}_\Sigma$ be the declared Hausdorff target space containing the regular states used before contact and the admissible target germs. A target-shell presentation consists of nested target neighborhoods $U_r(\Sigma) \subset \hat{X}_\Sigma$, $r \downarrow 0$, and normalized observables $\Phi_r : \hat{X}_\Sigma \rightarrow [0, 1]$ with:

(H1) $U_r(\Sigma) \subseteq U_{2r}(\Sigma)$ and $\bigcap_{r>0} U_{2r}(\Sigma) = \Sigma$ in the declared state or germ topology;

(H2)

$$\Phi_r = 0 \quad \text{on } \hat{X}_\Sigma \setminus U_{2r}(\Sigma), \quad \Phi_r = 1 \quad \text{on } U_r(\Sigma); \quad (20)$$

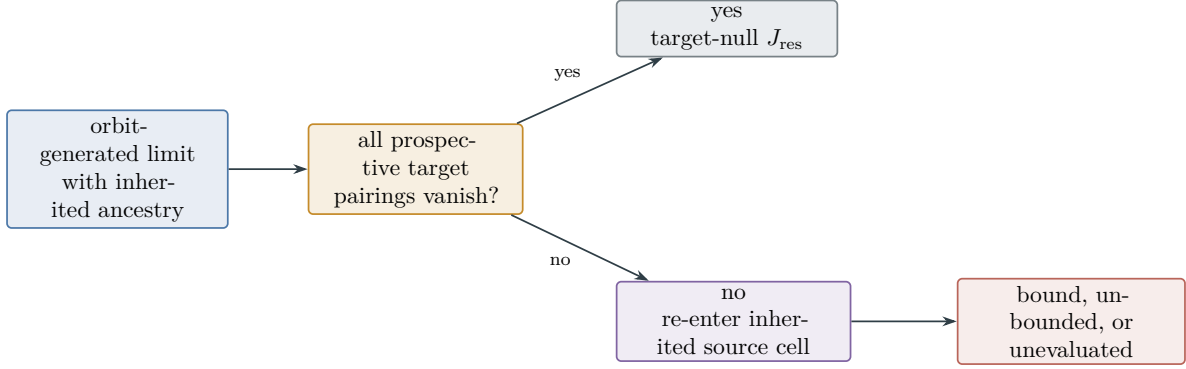


Figure 7: Target-visible limits retain their inherited source cell. The only residual output is target-null. A nonzero pairing fixes structural ancestry before any attempt is made to estimate the corresponding source profile.

(H3) Φ_r is continuous in the orbit topology, is differentiable along every regular pre-contact segment on which it is used, and $\Phi_r(u(\cdot))$ is absolutely continuous there with

$$\frac{d}{dt}\Phi_r(u_t) = \langle D\Phi_r(u_t), J_t \rangle \quad \text{for a.e. } t; \quad (21)$$

(H4) the current functional $J \mapsto \langle D\Phi_r(z), J \rangle$ belongs to Γ_Σ for every pre-contact shell state z . The contact time is $\tau_\Sigma(u) = \inf\{t : u(t) \in \Sigma\}$, with the germ interpretation when $T_*(u)$ is a noncontinuation time. If $u(0) \notin \Sigma$ and u contacts Σ , then for every sufficiently small r there are strictly pre-contact times

$$0 \leq t_r^- < t_r^+ < \tau_\Sigma(u), \quad u(t_r^-) \notin U_{2r}(\Sigma), \quad u(t_r^+) \in U_r(\Sigma). \quad (22)$$

Indeed, the first two shell conditions separate $u(0)$ from Σ at small scales, while contact supplies the second time. For a target at infinity, the same definition is imposed in the declared compactification or normalization chart; chart formation retains its **Abs/Lift** ancestry.

Theorem I.5.2 (Normalized shell-crossing accountability). *Let u be a legal orbit generated by the compiled continuum algebra, with $u(0) \notin \Sigma$, and suppose u contacts Σ . For every sufficiently small r , choose a crossing interval $I_r = [t_r^-, t_r^+]$ as in (22). Write the actual physical current as*

$$J = \sum_{c \in \mathfrak{C}} J^c + J^{\text{res}},$$

where $J^{\text{res}} \in \mathcal{N}_\Sigma$, and assume all six shell pairings are integrable on I_r . Define the signed source progress

$$Q_{c,r}(u) := \int_{I_r} \langle D\Phi_r(u_t), J_t^c \rangle dt. \quad (23)$$

Then the shell crossing has the exact accountability identity

$$1 = \Phi_r(u(t_r^+)) - \Phi_r(u(t_r^-)) = \sum_{c \in \mathfrak{C}} Q_{c,r}(u), \quad (24)$$

because

$$\int_{I_r} \langle D\Phi_r(u_t), J_t^{\text{res}} \rangle dt = 0. \quad (25)$$

Consequently

$$\sum_{c \in \mathfrak{C}} (Q_{c,r}(u))_+ \geq 1, \quad \max_{c \in \mathfrak{C}} (Q_{c,r}(u))_+ \geq \frac{1}{5}. \quad (26)$$

Equivalently,

$$\boxed{u \text{ contacts } \Sigma \implies \forall r \text{ sufficiently small } \exists c \in \mathfrak{C} : Q_{c,r}(u) > 0.} \quad (27)$$

Thus contact itself exposes a target-aligned pre-contact occurrence in an existing source cell at every sufficiently small shell.

Proof. The endpoint normalization (20) gives the first equality in (24). Absolute continuity and the chain rule give

$$1 = \int_{I_r} \langle D\Phi_r(u_t), J_t \rangle dt.$$

Insert the current normal form proved by Theorem I.4.4. Each shell derivative is a member of Γ_Σ , so $J^{\text{res}} \in \mathcal{N}_\Sigma$ proves (25) and leaves the sum of the five quantities in (23). Since $\sum_c x_c \leq \sum_c (x_c)_+$, their positive parts sum to at least one. One of five nonnegative numbers whose sum is at least one is at least $1/5$. $\square$

Corollary I.5.3 (No spontaneous deterministic contact). *Let u be a legal orbit generated by the compiled continuum algebra, with $u(0) \notin \Sigma$, and let the target carry a normalized shell presentation. Suppose every source current is target-null along every pre-contact shell:*

$$\langle D\Phi_r(u_t), J_t^c \rangle = 0 \quad \text{for a.e. } t < \tau_\Sigma(u), \quad c \in \mathfrak{C}. \quad (28)$$

Then every $t \mapsto \Phi_r(u_t)$ is constant on its regular pre-contact interval, and u cannot contact Σ . Target avoidance also follows when every shell integral $Q_{c,r}(u)$ is nonpositive. For a singularity at infinity, these statements apply to the normalized shell current in the declared compactification.

Proof. The chain rule, compiled normal form, residual target-nullity, and (28) give

$$\Phi_r(u_t) - \Phi_r(u_s) = 0$$

on every regular pre-contact interval $[s, t]$. Hence each shell observable is constant. Contact would instead give $1 = \sum_c Q_{c,r}(u)$ for every sufficiently small shell. Under the weaker nonpositivity hypothesis, the right side is at most zero. Both cases contradict the unit crossing. $\square$

Corollary I.5.4 (Normalized contact at infinity). *Suppose singular contact is defined only after legal normalizations $v_r = \mathcal{S}_r u$ into a state or germ compactification carrying normalized shells $\tilde{\Phi}_r$. If the normalized current $\tilde{J}_r$ has the compiled normal form, including every chart derivative with its Abs/Lift ancestry, then every normalized crossing interval satisfies*

$$1 = \sum_{c \in \mathfrak{C}} \int_{I_r} \langle D\tilde{\Phi}_r(v_r(t)), \tilde{J}_{r,t}^c \rangle dt, \quad \max_{c \in \mathfrak{C}} \left(\int_{I_r} \langle D\tilde{\Phi}_r(v_r(t)), \tilde{J}_{r,t}^c \rangle dt \right)_+ \geq \frac{1}{5}.$$

Thus blow-up at infinity cannot occur without target-aligned structure in an existing source cell.

Proof. Apply Theorem I.5.2 in the normalized state space. Source preservation under the legal chart is Theorem I.3.4. $\square$

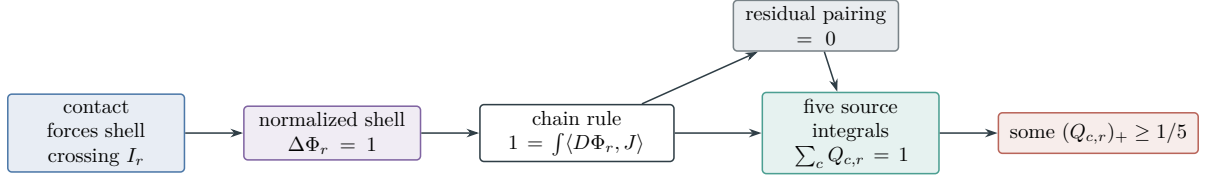


Figure 8: Normalized shell-crossing accountability. Contact forces a unit increase of the shell observable along the actual pre-contact current. The target-null residual contributes zero, so one of the five source integrals has positive part at least $1/5$.

Theorem I.5.5 (Resolvent source-response comparison). *Let L and L_0 be closed killed generators on the same Banach space, and let $\lambda \in \rho(L) \cap \rho(L_0)$. Write*

$$R_L(\lambda) = (\lambda - L)^{-1}, \quad R_0(\lambda) = (\lambda - L_0)^{-1}.$$

Assume $R_0(\lambda)k_\Sigma \in D(L) \cap D(L_0)$, and suppose

$$L - L_0 = \sum_{c \in \mathfrak{C}} K^c + K^{J_{\text{res}}} \quad (29)$$

on that vector. Then, for every continuous initial functional ℓ_0 ,

$$\begin{aligned} \langle \ell_0, (R_L(\lambda) - R_0(\lambda))k_\Sigma \rangle &= \sum_{c \in \mathfrak{C}} \langle R_L(\lambda)^* \ell_0, K^c R_0(\lambda)k_\Sigma \rangle \\ &\quad + \langle R_L(\lambda)^* \ell_0, K^{J_{\text{res}}} R_0(\lambda)k_\Sigma \rangle. \end{aligned} \quad (30)$$

If the last pairing is target-null, it vanishes. If the real part p of the left side is positive, some existing source cell has real pairing at least $p/5$.

Proof. The second resolvent identity gives

$$R_L(\lambda) - R_0(\lambda) = R_L(\lambda)(L - L_0)R_0(\lambda).$$

Pair with ℓ_0 and k_Σ , then insert (29). Target-nullity removes the residual term. A positive sum of five real numbers has a summand at least one fifth of that sum. $\square$

Remark I.5.6 (Arrival semantics). Equation (30) is an operator identity. It becomes a difference of discounted contact responses when the killed semigroups admit the target-flux realization of Theorem II.4.4. No probabilistic meaning is inferred from the algebraic resolvent alone.

Theorem I.5.7 (Stochastic no-spontaneous-contact accountability). *Let τ_Σ be a stopping time and $N_t = \mathbf{1}_{\{\tau_\Sigma \leq t\}}$. Suppose $N - A$ is a martingale and the predictable compensator on $[0, T]$ has the finite-variation decomposition*

$$A = \sum_{c \in \mathfrak{C}} A^c + A^{J_{\text{res}}}, \quad (31)$$

where all measures are supported on $\{t < \tau_\Sigma\}$, the signed source terms are integrable, every term starts from zero, $\mathbb{P}\{\tau_\Sigma = 0\} = 0$, and $\mathbb{E}A_T^{J_{\text{res}}} = 0$. Then

$$\mathbb{P}\{\tau_\Sigma \leq T\} = \sum_{c \in \mathfrak{C}} \mathbb{E}A_T^c. \quad (32)$$

If $p_T := \mathbb{P}\{\tau_\Sigma \leq T\} > 0$, one pre-contact source compensator satisfies $\mathbb{E}A_T^c \geq p_T/5$. In particular, all five source expectations cannot vanish when contact has positive probability.

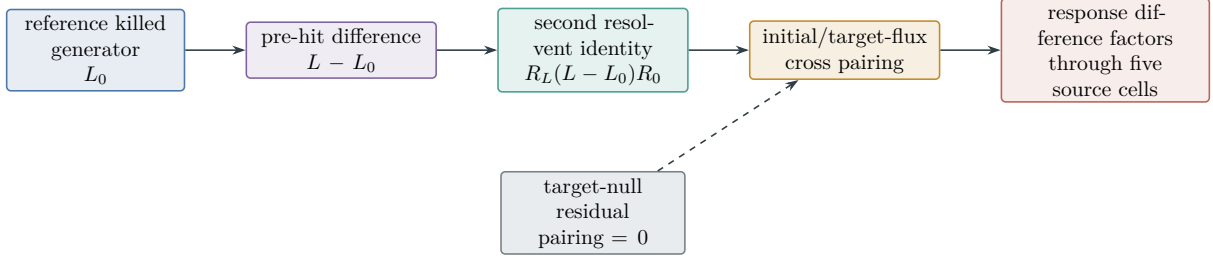


Figure 9: Resolvent source-response comparison. A discounted target-response difference is sandwiched around the pre-hit generator difference and therefore inherits its five-source decomposition. The no-spontaneous-contact theorem itself comes from normalized shell crossing.

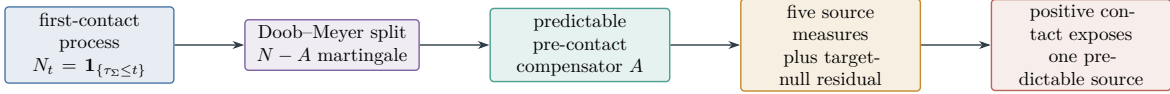


Figure 10: Stochastic no-spontaneous-contact accountability. First contact is charged by its predictable pre-contact compensator. A positive contact probability forces a positive contribution from one of the five source measures before the stopping time.

Proof. Taking expectations of the integrable martingale $N - A$, using the stated zero initial values, gives $\mathbb{E}N_T = \mathbb{E}A_T$. Insert (31) and use the zero expectation of the residual term. The final assertion follows from the five-term pigeonhole argument. $\square$

Corollary I.5.8 (Continuum prospective source exhaustion). *Normalized deterministic contact and positive-probability stochastic contact both force a positive pre-contact occurrence with ancestry in $\mathfrak{C}$. A positive killed-resolvent response difference has the same source-extraction property. The post-saturation residual contribution is zero in all three interfaces.*

Definition I.5.9 (Saturated prospective source profile). Fix one of the three prospective interfaces, restricted to records satisfying its stated target-null residual condition, and fix its normalization. Write $\text{Prog}_\Sigma^c(R)$ for the signed c -source contribution: $Q_{c,r}$ for a deterministic shell, $\mathbb{E}A_T^c$ for a stochastic contact record, and the real c -pairing in (30) for a resolvent comparison. Set

$$\text{Prog}_\Sigma(R) := \sum_{c \in \mathfrak{C}} \text{Prog}_\Sigma^c(R). \quad (33)$$

Define

$$\mathsf{T}_{\Sigma,c}^{\text{phys,sat}}(B_{\text{phys}}) := \sup_{R \in \mathcal{R}^{\text{phys,sat}}(B_{\text{phys}})} (\text{Prog}_\Sigma^c(R))_+, \quad (34)$$

$$\mathsf{T}_\Sigma^{\text{phys,sat}}(B_{\text{phys}}) := \sum_{c \in \mathfrak{C}} \mathsf{T}_{\Sigma,c}^{\text{phys,sat}}(B_{\text{phys}}). \quad (35)$$

The supremum is restricted to records on which the selected interface is defined; it may take the value $+\infty$. An *unevaluated* mark is metadata about the availability of an upper bound, not an additional value of this profile.

Theorem I.5.10 (Source-only prospective envelope). *Every record in the domain of a prospective interface satisfies*

$$(\text{Prog}_\Sigma(R))_+ \leq \mathsf{T}_\Sigma^{\text{phys,sat}}(B_{\text{phys}}). \quad (36)$$

The envelope consists solely of the five source profiles. Any finite upper bounds on those profiles give a complete prospective progress bound. A normalized deterministic contact record satisfies $\text{Prog}_\Sigma(R) = 1$.

Proof. Each of Theorems I.5.2, I.5.5 and I.5.7 has, after removal of its target-null residual pairing, the form

$$\text{Prog}_\Sigma(R) = \sum_{c \in \mathfrak{C}} \text{Prog}_\Sigma^c(R).$$

Take the positive part, bound every signed summand by its positive part, and then apply (34). $\square$

Definition I.5.11 (Singularity-aligned structural support). For a pre-contact record R in any prospective interface, define

$$\text{Align}_\Sigma(R) := \{c \in \mathfrak{C} : \text{Prog}_\Sigma^c(R) > 0\}. \quad (37)$$

Equivalently, in the deterministic interface an occurrence is aligned when the singularity-descending covector $D\Phi_r(u_t)$ has positive integrated pairing with its source current on the crossing interval. The word “aligned” is target-relative: it does not rename a source as **Caus**, and it is evaluated only along the fixed equation.

Theorem I.5.12 (No-spontaneous-progress principle). *For every prospective record,*

$$\text{Prog}_\Sigma(R) > 0 \implies \text{Align}_\Sigma(R) \neq \emptyset, \quad \max_{c \in \mathfrak{C}} (\text{Prog}_\Sigma^c(R))_+ \geq \frac{\text{Prog}_\Sigma(R)}{5}. \quad (38)$$

If every source occurrence represented by R lies in $\mathcal{N}_\Sigma$, then

$$\text{Prog}_\Sigma(R) = 0, \quad (39)$$

regardless of the size, oscillation, or compactness of those target-null currents. More generally, if every source contribution is nonpositive, then positive target progress is impossible. For a normalized deterministic shell crossing, $\text{Prog}_\Sigma(R) = 1$, so contact yields $\max_c (\text{Prog}_\Sigma^c(R))_+ \geq 1/5$.

Proof. Equation (33) is the compiled prospective identity. A positive sum has a positive summand, and its largest positive part is at least one fifth of the sum. If every source occurrence lies in the common kernel of the prospective target family, every summand is zero; this proves (39). The final shell assertion is Theorem I.5.2. $\square$

Corollary I.5.13 (No-spontaneous-contact regularity). *Let the target have a normalized shell presentation, and let every legal orbit be generated by the compiled algebra on its regular pre-contact interval. If $\text{Align}_\Sigma(R) = \emptyset$ for every legal shell record, then the target is unreachable. In particular, if all five source currents have zero class in $\mathcal{J}_\Sigma$ along every pre-contact shell, their ancestry is retained while their target contribution lies in $\mathcal{N}_\Sigma$, and contact is impossible:*

$$\boxed{q_\Sigma(J^c) = 0 \text{ on every pre-contact target shell for all } c \in \mathfrak{C} \implies \tau_\Sigma = \infty.} \quad (40)$$

This conclusion requires neither compactness of the full orbit nor a positive physical-price estimate.

Proof. If an orbit contacted Σ , Theorem I.5.2 would provide a normalized shell record with unit progress. By Theorem I.5.12, its aligned support would be nonempty, contrary to the hypothesis. Under the displayed quotient condition, every shell source integral vanishes, so the same contradiction follows directly from (24). $\square$

Remark I.5.14 (Why the target quotient is stronger than a full norm bound). A classical energy estimate commonly controls every component of a current, including directions irrelevant to the singular target. The quotient $\mathcal{J}_\Sigma$ discards a current only after all declared prospective target pairings vanish. The discarded current may be large or noncompact; no estimate of it is needed for (39). Only the target-aligned quotient directions proceed to resistance, compactness, rigidity, or physical-price certificates. The quotient strictly reduces the regularity problem because target-null directions are absent from all downstream estimates.

Part II

Physical Certificate Calculus

II.1 Structural invariant reduction of analytic certificates

The closed source algebra separates the origin of a continuum quantity from the estimate used to control it. Once a source family has been placed in a closed target-null quotient or removed from the admissible orbit family by a proved physical exclusion, later certificates are evaluated on the remaining reduced space. The analytic calculation then has only three possible jobs: close a remaining class, compare two nonnegative forms, or classify the equality set of a sharp comparison.

Definition II.1.1 (Certificate-compatible reduction). Let Y be a Hausdorff locally convex space of currents, profiles, or limit records, and let $K \subseteq Y$ be a closed linear subspace already annihilated by the downstream target and certificate maps. Write

$$\pi_K : Y \longrightarrow Y/K.$$

A downstream map, form, compactification, or source transport is *compatible with the reduction* when it is constant on cosets of K and therefore factors continuously through π_K . A previously excluded closed family is removed from the admissible domain before the quotient is formed. Thus a reduction never discards a target-visible family without the certificate that excludes that family.

Theorem II.1.2 (Propagation of certified reductions). *Let $K_0 \subseteq K_1 \subseteq Y$ be closed subspaces, and assume every later certificate is compatible with both reductions. Then the canonical map is a topological linear isomorphism*

$$(Y/K_0)/(K_1/K_0) \cong Y/K_1. \quad (41)$$

Consequently a source direction closed at one stage may be propagated into the next quotient and omitted from every later compatible calculation. A later invariant computed after reduction has the same value as the corresponding invariant on Y after quotienting by all previously closed directions.

Proof. The quotient map $Y/K_0 \rightarrow Y/K_1$ is continuous, open, and has kernel K_1/K_0 . The topological first isomorphism theorem gives (41). Every compatible downstream object factors through these quotient maps, so its reduced and unreduced evaluations agree under the displayed isomorphism. $\square$

Theorem II.1.3 (Target-quotient coercivity). *Let Y be a Hilbert source space, let $K \subseteq Y$ be a closed target-null subspace, and let $T : Y \rightarrow Z_\Sigma$ be bounded with $K \subseteq \ker T$. Let $q : Y \rightarrow [0, \infty]$ be an extended nonnegative quadratic form with finite domain $D(q)$ satisfying $D(q) + K \subseteq D(q)$, and define its relaxed quotient form by*

$$q_{\text{red}}([y]) := \inf_{k \in K} q(y + k), \quad [y] \in Y/K. \quad (42)$$

If, on the required reduced domain,

$$\|[y]\|_{Y/K}^2 \leq C q_{\text{red}}([y]), \quad (43)$$

then

$$\|Ty\|_{Z_\Sigma}^2 \leq \|\tilde{T}\|^2 C q_{\text{red}}([y]) \leq \|\tilde{T}\|^2 C q(y), \quad (44)$$

where $\tilde{T} : Y/K \rightarrow Z_\Sigma$ is the descended target map. No bound for the full norm $\|y\|_Y$ follows or is needed.

Proof. Because T annihilates K , the quotient universal property gives the bounded map $\tilde{T}$ with $T = \tilde{T}\pi_K$. Hence

$$\|Ty\| \leq \|\tilde{T}\| \|[y]\|_{Y/K}.$$

Apply (43); the final inequality follows by using $k = 0$ in (42). $\square$

Example II.1.4 (Strict gain over full-state coercivity). Let $Y = K \oplus Z_\Sigma$, let $T(k, z) = z$, and let $q(k, z) = \|z\|^2$. Then (43) holds with $C = 1$, while no inequality $\|(k, z)\|_Y^2 \leq C' q(k, z)$ can hold when $K \neq \{0\}$. The sequence $(nk_0, 0)$ may diverge in the full state norm but is exactly target-null. Thus the quotient theorem proves every target estimate available from q without controlling that divergent motion.

Definition II.1.5 (Structural certificate invariants). Work on a certificate-compatible reduced space Y_{red} , with a continuous target projection $\Pi_\Sigma : Y_{\text{red}} \rightarrow Z_\Sigma$. The structural invariants used below have the following forms.

(I1) For a candidate family $\mathcal{F} \subseteq Y_{\text{red}}$, its *target closure invariant* is

$$\mathfrak{H}_{\mathcal{F}}^\Sigma := \overline{\text{span } \Pi_\Sigma(\mathcal{F})}^{Z_\Sigma}. \quad (45)$$

(I2) For nonnegative functionals D, N_+ on a reduced admissible family $\mathcal{P}$, the *sharp domination radius* is

$$\alpha_*(N_+ \mid D) := \sup_{\substack{V \in \mathcal{P} \\ D(V) > 0}} \frac{N_+(V)}{D(V)}. \quad (46)$$

The value is $+\infty$ when some V has $D(V) = 0 < N_+(V)$. For a normalized family $\mathcal{N} \subseteq \mathcal{P}$, the *physical-price gap* is

$$\delta_*(D \mid \mathcal{N}) := \inf_{V \in \mathcal{N}} D(V). \quad (47)$$

(I3) Suppose $\alpha_* < \infty$, and let $\hat{\mathcal{E}}_{\alpha_*}$ be a declared compactification of the normalized equality family

$$\mathcal{E}_{\alpha_*} := \{V \in \mathcal{P} : D(V) > 0, N_+(V) = \alpha_* D(V)\}.$$

If the equality identities determine a continuous remainder map $\mathcal{R}_{\text{eq}} : \hat{\mathcal{E}}_{\alpha_*} \rightarrow Y_{\text{red}}$, the *equality-rigidity invariant* is

$$\mathfrak{H}_{\text{eq}}^\Sigma := \overline{\text{span } \Pi_\Sigma \mathcal{R}_{\text{eq}}(\hat{\mathcal{E}}_{\alpha_*})}^{Z_\Sigma}. \quad (48)$$

The equality identities may be saturation equations, conservation laws, modulation equations, Pohozaev identities, or Liouville constraints.

Each invariant is computed only after all compatible source, gauge, trace, and target-null reductions already proved have been propagated.

Theorem II.1.6 (Structural normal form for the certificate calculus). *Within the interfaces of this paper, every analytic input to target exclusion has one of the following exact structural meanings.*

- (N1) *A candidate family is target-null on the reduced interface exactly when $\mathfrak{H}_{\mathcal{F}}^{\Sigma} = 0$.*
- (N2) *The domination inequality $N_+ \leq aD$ holds on $\mathcal{P}$ exactly when $\alpha_*(N_+ \mid D) \leq a$.*
- (N3) *The normalized family $\mathcal{N}$ has a uniform positive physical price exactly when $\delta_*(D \mid \mathcal{N}) > 0$.*
- (N4) *A sharp equality branch has no target-visible remainder exactly when $\mathfrak{H}_{\text{eq}}^{\Sigma} = 0$.*

Source closure and target-null residual calculations use (N1); resolvent, flux, and carrier estimates use (N2); compactness–rigidity and cascade prices use (N3); sharp or critical cases use (N4). Thus an explicit analytic estimate enters the continuum calculus only through a quotient, a sharp radius or gap, or an equality remainder computed on the propagated reduced source space.

Proof. Item (N1) is the definition of the closed target projection in (45). For (N2), divide by $D(V)$ when it is positive; the convention for a positive D -null mode gives the remaining case. Item (N3) is the definition of the infimum in (47). Item (N4) follows from (48): its vanishing is precisely the vanishing of every target projection of the equality remainder. The final classification lists the four analytic obligations occurring in Theorems I.4.5 and I.5.10, Definition II.7.2, and Lemma II.7.3. $\square$

Proposition II.1.7 (Positive separator for an equality branch). *Assume that every $Z \in \hat{\mathcal{E}}_{\alpha_*}$ satisfies an identity*

$$0 = Q_{\Sigma}(\Pi_{\Sigma}\mathcal{R}_{\text{eq}}(Z)) + \sum_{j=1}^m q_j(Z), \quad (49)$$

where $Q_{\Sigma} \geq 0$ is positive definite on Z_{Σ} and every $q_j \geq 0$. Then $\mathfrak{H}_{\text{eq}}^{\Sigma} = 0$.

Proof. Every term on the right of (49) is nonnegative, so each term vanishes. Positive definiteness gives $\Pi_{\Sigma}\mathcal{R}_{\text{eq}}(Z) = 0$ for every Z . Taking the closed linear span proves the claim. $\square$

II.2 Covariant action and the physical budget

The fixed-Caus formulation separates the current selected by the PDE from the metric used to measure irreversible expenditure. This separation is physical: the action below is the quantity appearing in the energy identity. The explicit inverse notation in this section is used only when the mobility has the Hilbert realization stated next. In a general Gelfand triple, the same identities are read at the level of the closed mobility form.

Definition II.2.1 (Mobility current space). Fix a state u , a real Hilbert tangent space $\mathcal{H}_u$, and a bounded self-adjoint nonnegative mobility $\mathbb{K}(u) \in \mathcal{L}(\mathcal{H}_u)$. Let $\mathbb{K}(u)^{-1/2}$ denote the Moore–Penrose inverse on $(\ker \mathbb{K}(u))^{\perp}$. The mobility current space is the Cameron–Martin space

$$\mathfrak{J}_u := \text{Ran } \mathbb{K}(u)^{1/2}$$

with inner product and norm

$$(J_1, J_2)_{u,*} := \langle \mathbb{K}(u)^{-1/2} J_1, \mathbb{K}(u)^{-1/2} J_2 \rangle_{\mathcal{H}_u}, \quad \|J\|_{u,*}^2 := \|\mathbb{K}(u)^{-1/2} J\|_{\mathcal{H}_u}^2.$$

Equivalently, this is the completion of $\text{Ran } \mathbb{K}(u)$ in the displayed norm. Covectors are represented in $\mathcal{H}_u$, with components in $\ker \mathbb{K}(u)$ identified when they act on currents. For a velocity v , put

$$W(u, v) := v + A_{\text{Caus}}(u) - f_{\text{Bdry}}(u).$$

The dissipation Lagrangian is

$$\mathcal{L}(u, v) := \begin{cases} \frac{1}{2} \|W(u, v)\|_{u,*}^2, & W(u, v) \in \mathfrak{J}_u, \\ +\infty, & \text{otherwise.} \end{cases} \quad (50)$$

Its physical action on an interval I is $\mathcal{A}_I(u) = \int_I 2\mathcal{L}(u, \dot{u}) \, dt$.

Definition II.2.2 (Covariant dissipation Hamiltonian). For $p \in \mathcal{H}_u$, define

$$\mathcal{H}_{\text{cov}}(u, p) := -\langle p, A_{\text{Caus}}(u) - f_{\text{Bdry}}(u) \rangle + \frac{1}{2} \|\mathbb{K}(u)^{1/2} p\|^2. \quad (51)$$

This is the Legendre–Fenchel dual of $v \mapsto \mathcal{L}(u, v)$.

Proposition II.2.3 (Fenchel equality and the selected current). *Whenever the displayed quantities are finite,*

$$\mathcal{L}(u, v) + \mathcal{H}_{\text{cov}}(u, p) - \langle p, v \rangle = \frac{1}{2} \|W(u, v) - \mathbb{K}(u)p\|_{u,*}^2 \geq 0. \quad (52)$$

Equality holds precisely when $W(u, v) = \mathbb{K}(u)p$ in the current quotient. For the fixed-Caus gradient flow (3), the selected velocity has

$$W(u, \dot{u}) = -\mathbb{K}(u)D\mathcal{E}(u),$$

so equality holds at $p = -D\mathcal{E}(u)$, modulo the mobility kernel.

Proof. Substitute (50) and (51); since $v = W - A_{\text{Caus}} + f_{\text{Bdry}}$, the linear terms combine to $-\langle p, W \rangle$. Completing the square in the $\mathbb{K}^{-1}$ -metric gives (52). The equality and selected-current statements follow directly. $\square$

Definition II.2.4 (Physical carrier balance). Let $u : [0, T_*) \rightarrow X$ be a legal orbit. A physical carrier balance consists of a scalar $\mathcal{K}_u : [0, T_*) \rightarrow \mathbb{R}$, bounded below by K_- , and nonnegative Radon measures $\mathfrak{D}_u, \Phi_u^+, R_u^+$ such that, for $0 \leq s < t < T_*$,

$$\mathcal{K}_u(t) + \mathfrak{D}_u((s, t]) \leq \mathcal{K}_u(s) + \Phi_u^+((s, t]) + R_u^+((s, t]). \quad (53)$$

Here $\mathfrak{D}_u$ is the gross physical expenditure, Φ_u^+ is the feeding flux returned to the active region, and R_u^+ is legitimate positive exterior input not included in that flux. A passivity certificate with constant $\beta \leq 1$ is the measure domination

$$\Phi_u^+ \leq \beta \mathfrak{D}_u. \quad (54)$$

All terms are expressed in the physical clock. A normalized-clock identity qualifies only after the comparison in Proposition II.9.5 has been verified.

Theorem II.2.5 (Carrier compiler for the single physical budget). *Assume Definition II.2.4, (54), and $R_u^+([0, T_*)) < \infty$. Then*

$$\mu_u := \mathfrak{D}_u - \Phi_u^+ \quad (55)$$

is a nonnegative countably additive physical expenditure measure and

$$\mu_u([0, T_*)) \leq \mathcal{K}_u(0) - K_- + R_u^+([0, T_*)) =: B_{\text{phys}} < \infty. \quad (56)$$

If $\beta < 1$, then

$$\mu_u \geq (1 - \beta)\mathfrak{D}_u. \quad (57)$$

For $\beta = 1$, μ_u is the exact flux-deficit measure. Any other nonnegative expenditure needed by the application may be incorporated only after a combined carrier balance of the form (53) has been proved for it. The combined measures are fixed before μ_u and the sole ceiling B_{phys} are declared; no independent budget is appended later.

Proof. The domination $\Phi_u^+ \leq \mathfrak{D}_u$ makes the difference in (55) a nonnegative Radon measure. Rearrange (53) with $s = 0$, use $\mathcal{K}_u(t) \geq K_-$, and let $t \uparrow T_*$. Monotone convergence gives (56). Finally, $\Phi_u^+ \leq \beta\mathfrak{D}_u$ gives (57). $\square$

Definition II.2.6 (Reduced passivity radius). Let $\mathcal{P}_{\text{car}}^{\text{red}}$ be a normalized carrier profile space obtained after certificate-compatible source reductions. Let $D(V) \geq 0$ and $\Phi_+(V) \geq 0$ be the descended gross-expenditure and feeding-flux densities in the same physical units. The reduced passivity radius is

$$\beta_*^{\text{red}} := \alpha_*(\Phi_+ \mid D) = \sup_{\substack{V \in \mathcal{P}_{\text{car}}^{\text{red}} \\ D(V) > 0}} \frac{\Phi_+(V)}{D(V)}, \quad (58)$$

with the null-mode convention of Definition II.1.5. A bound $\beta_*^{\text{red}} \leq 1$ supplies (54) whenever the carrier identity is local in the normalized profiles and its physical-clock pullback is proved.

Proposition II.2.7 (Quadratic passivity is a spectral invariant). *Suppose on the reduced Hilbert profile space*

$$D(V) = \langle AV, V \rangle, \quad \Phi_+(V) = \max\{\langle BV, V \rangle, 0\},$$

where $A \geq 0$, $B = B^$, all D -null target-visible feeding modes have been separated, and B is bounded in the A -form norm. Let $S_{\text{flux}} = A^{-1/2}BA^{-1/2}$ be the descended self-adjoint form operator on the completion modulo $\ker A$. Then*

$$\beta_*^{\text{red}} = \max\{\sup \sigma(S_{\text{flux}}), 0\}. \quad (59)$$

Thus every multiplier or casewise proof of $\Phi_+ \leq D$ is, on this reduced quadratic interface, an upper bound for one spectral radius.

Proof. The A -form completion is a Hilbert space after quotienting by $\ker A$. The form induced by B is represented there by S_{flux} . Rayleigh–Ritz identifies the supremum of $\langle BV, V \rangle / D(V)$ with $\sup \sigma(S_{\text{flux}})$; taking the positive part gives (59). $\square$

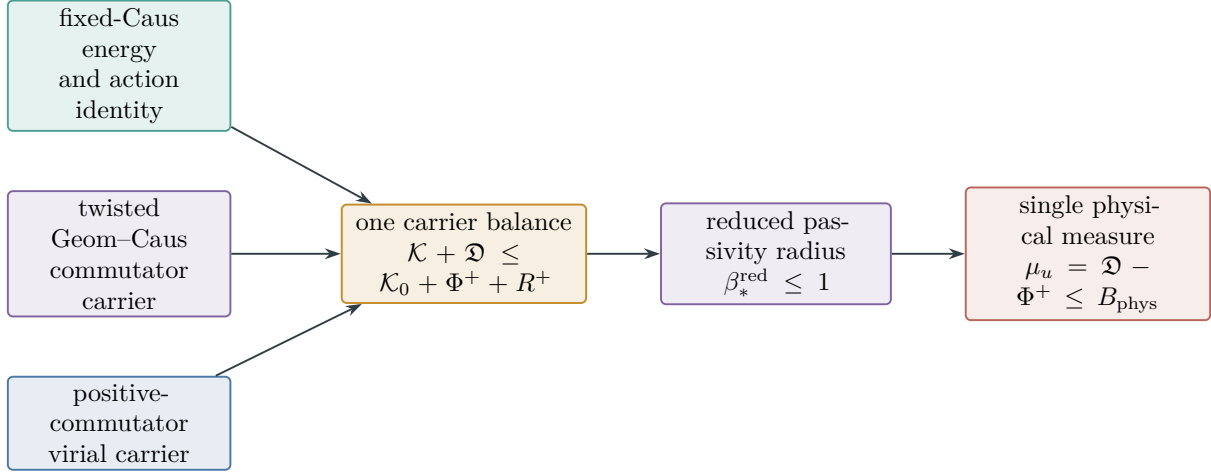


Figure 11: Carrier-to-budget compiler. Gradient-flow, hypocoercive, and positive-commutator identities are different ways to expose the same structural object: a lower-bounded carrier whose feeding flux is dominated by gross expenditure. The output is the sole physical budget measure.

II.3 Hamiltonian meanings and chart covariance

Definition II.3.1 (Geometric spectral Hamiltonian). Let $\mathfrak{q}$ be a densely defined closed nonnegative symmetric form on a real Hilbert space $\mathcal{H}$. Its geometric Hamiltonian is the unique self-adjoint operator $H \geq 0$ satisfying

$$\mathfrak{q}(u, v) = \langle H^{1/2}u, H^{1/2}v \rangle, \quad u, v \in D(\mathfrak{q}).$$

This Hamiltonian records only the **Geom** form; the fixed antisymmetric current remains in **Caus**.

Proposition II.3.2 (Separation of Hamiltonian meanings). *The following constructions require distinct data.*

- (i) A closed symmetric form determines the spectral Hamiltonian H of Definition II.3.1.
- (ii) A canonical conservative Hamiltonian vector field requires a Poisson tensor $\mathbb{J}$, a domain on which the Jacobi identity holds, and a functional $\mathcal{H}$; the field is $\mathbb{J}D\mathcal{H}$.
- (iii) A positive mobility and a covariant connection determine the Fenchel Hamiltonian $\mathcal{H}_{\text{cov}}$ of Definition II.2.2.

An antisymmetric **Caus** form alone supplies directed transport but does not provide the Poisson or Jacobi data in item (ii).

Proof. The representation theorem for closed nonnegative forms gives item (i) [5, Chapter VI]. Item (ii) is the definition of a Poisson Hamiltonian system; antisymmetry is only one of the Poisson requirements. The square completion in Proposition II.2.3 proves item (iii). $\square$

Theorem II.3.3 (Same-current covariance). *Let $v = \Phi_t(u)$ be a C^1 family of local diffeomorphisms between the relevant Hilbert charts. Assume $T_t = D\Phi_t(u) : \mathcal{H}_u \rightarrow \mathcal{H}_v$ is a bounded linear isomorphism with bounded inverse. With every expression on the right evaluated at $u = \Phi_t^{-1}(v)$, define*

$$\tilde{\mathbb{K}} = T_t \mathbb{K} T_t^*, \quad \tilde{A}_{\text{Caus}} = T_t A_{\text{Caus}}, \quad A_{\text{Lift}}^\Phi = -\partial_t \Phi_t, \quad \tilde{f}_{\text{Bdry}} = T_t f_{\text{Bdry}}. \quad (60)$$

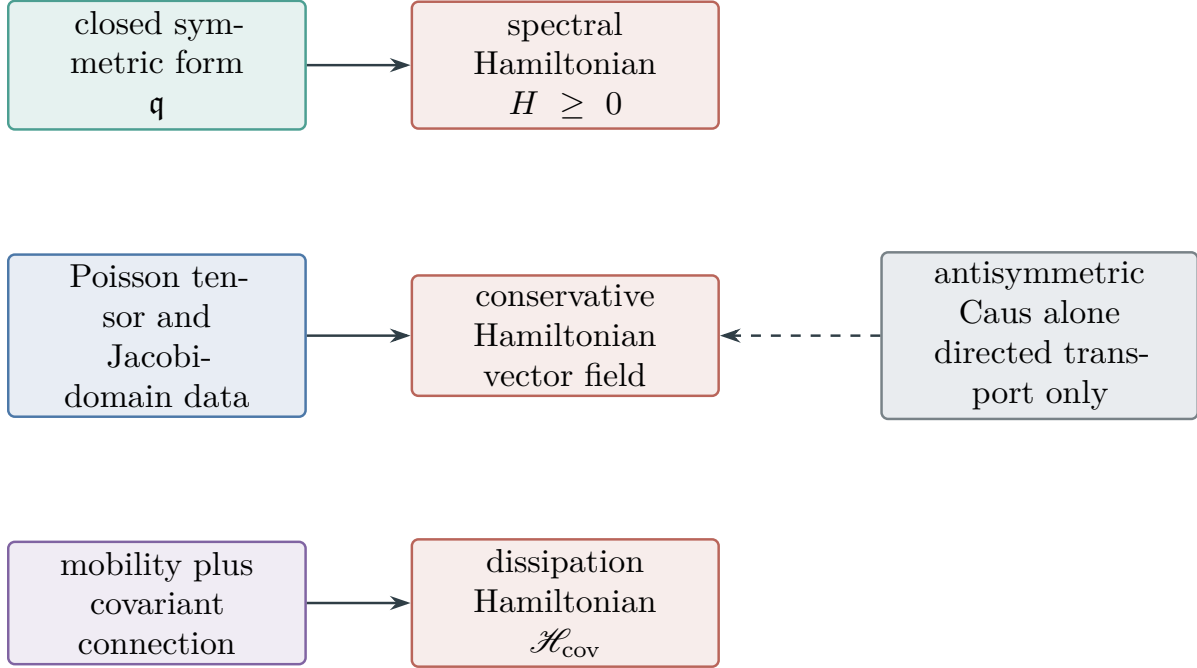


Figure 12: Three noninterchangeable Hamiltonian constructions. The dashed line marks a missing implication: skew transport alone does not establish a Poisson structure.

Then

$$\dot{v} + \tilde{A}_{\text{Caus}}(v) + A_{\text{Lift}}^{\Phi}(v) - \tilde{f}_{\text{Bdry}}(v) = T_t(\dot{u} + A_{\text{Caus}}(u) - f_{\text{Bdry}}(u)). \quad (61)$$

The map T_t is an isometry between the corresponding mobility-current spaces equipped with their transported metrics, so the Lagrangian and physical action are invariant. The chart has Lift ancestry, and quotienting any declared symmetry additionally has Abs ancestry.

Proof. Differentiate $v = \Phi_t(u)$ and substitute (60); the $\partial_t \Phi_t$ terms cancel, giving (61). For $\tilde{J} = T_t J$, the variational identity

$$\|J\|_{\mathbb{K}^{-1}}^2 = \sup_{p \in \mathcal{H}_u} \{2\langle p, J \rangle - \langle p, \mathbb{K}p \rangle\}$$

and the bijective change of covector $p = T_t^* \tilde{p}$ give $\|\tilde{J}\|_{\tilde{\mathbb{K}}^{-1}} = \|J\|_{\mathbb{K}^{-1}}$. Integrating proves action invariance. The source labels follow from Definition I.2.1. $\square$

Proposition II.3.4 (Chart covariance of the dissipation Hamiltonian). *Under the hypotheses of Theorem II.3.3, let $p = T_t^* \tilde{p}$, and define the transformed Hamiltonian using the total connection $\tilde{A}_{\text{Caus}} + A_{\text{Lift}}^{\Phi}$. Then*

$$\tilde{\mathcal{H}}_{\text{cov}}(t, v, \tilde{p}) = \mathcal{H}_{\text{cov}}(u, p) + \langle \tilde{p}, \partial_t \Phi_t(u) \rangle. \quad (62)$$

The additional term is the time-dependent Lift connection. It is not a new physical Caus term.

Proof. Insert (60) into (51). The quadratic term is invariant because $\tilde{\mathbb{K}} = T_t \mathbb{K} T_t^*$, while

$$-\langle \tilde{p}, T_t(A_{\text{Caus}} - f_{\text{Bdry}}) - \partial_t \Phi_t \rangle = -\langle p, A_{\text{Caus}} - f_{\text{Bdry}} \rangle + \langle \tilde{p}, \partial_t \Phi_t \rangle.$$

$\square$

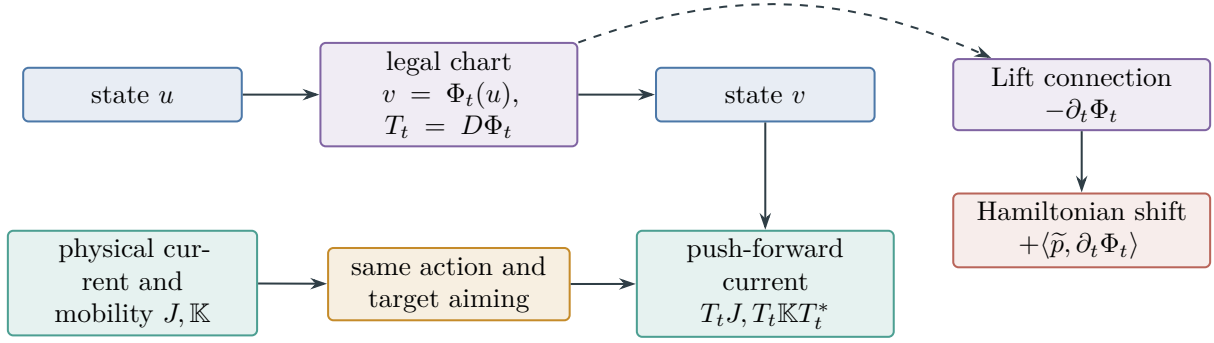


Figure 13: Same-current chart covariance. A chart transports the physical current and metric; its time dependence is a Lift connection with the exact Hamiltonian shift shown.

Remark II.3.5 (A change of physical clock is additional data). Theorem II.3.3 keeps the physical time variable fixed. If a blow-up chart also introduces $\tau = \theta(t)$, the current, mobility, and action density must be transformed with the exact Jacobian $dt/d\tau$. Action invariance, or comparison with the original physical measure, is not automatic; the required one-sided comparison is the content of Proposition II.9.5.

II.4 Killed resolvents, resistance, and target realization

Definition II.4.1 (Killed geometric Hamiltonian). Let $\mathfrak{q}$ be a densely defined closed nonnegative symmetric Markovian form and let Σ be quasi-closed for a regular or quasi-regular realization. The killed form is the restriction

$$D(\mathfrak{q}_\Sigma) = \{u \in D(\mathfrak{q}) : \tilde{u} = 0 \text{ q.e. on } \Sigma\}, \quad \mathfrak{q}_\Sigma = \mathfrak{q}|_{D(\mathfrak{q}_\Sigma)},$$

and its self-adjoint operator, when the restricted form is closed, is $H_\Sigma \geq 0$. Adding an admitted nonnegative killing potential instead defines a *soft*-killed form. It is a separate construction unless convergence to the hard restriction is proved.

Proposition II.4.2 (Existence of the killed Hamiltonian). *If $D(\mathfrak{q}_\Sigma)$ is closed in the form norm, then $\mathfrak{q}_\Sigma$ is closed and determines a unique self-adjoint $H_\Sigma \geq 0$.*

Proof. A form-norm closed subspace of the domain of a closed form is complete in that norm. The representation theorem for closed nonnegative forms gives H_Σ . $\square$

Proposition II.4.3 (Committor is the canonical binary-contact quotient). *Let A and B be disjoint quasi-closed sets for an associated right process, and let*

$$h(x) := \mathbb{P}_x\{\tau_A < \tau_B\}$$

be its quasi-continuous committor. Define $q_h(x) = h(x)$ outside the exceptional set. If a quotient $q : X \rightarrow Q$ preserves this binary contact law in the precise sense that $h = \bar{h} \circ q$ quasi-everywhere for some $\bar{h} : Q \rightarrow [0, 1]$, then

$$q_h = \bar{h} \circ q \quad \text{quasi-everywhere.} \quad (63)$$

Hence q_h is the coarsest quotient preserving the scalar event $\{\tau_A < \tau_B\}$: every other such quotient is finer than its level-set partition. This assertion concerns the binary hitting probability, not the full hitting-time distribution.

Proof. The preservation hypothesis is exactly (63). Therefore $q(x) = q(y)$ implies $h(x) = h(y)$, so every fiber of q lies in a fiber of q_h . Conversely, q_h itself preserves the event by the defining identity for h . $\square$

Theorem II.4.4 (Killed-resolvent stopwatch). *Let L_Σ generate a killed semigroup P_t^Σ with growth bound ω . Let ℓ_0 be an initial functional and let k_Σ be a terminal-flux vector such that*

$$\mathbb{P}_{\ell_0}\{\tau_\Sigma \in dt, \tau_\Sigma < \infty\} = \langle \ell_0, P_t^\Sigma k_\Sigma \rangle dt \quad (64)$$

as finite measures. Then, for $\lambda > \omega$,

$$\mathbb{E}_{\ell_0}[e^{-\lambda\tau_\Sigma}; \tau_\Sigma < \infty] = \langle \ell_0, (\lambda - L_\Sigma)^{-1} k_\Sigma \rangle. \quad (65)$$

Without (64), the right side remains a cross-resolvent pairing and has no automatic first-contact interpretation.

Proof. Integrate (64) against $e^{-\lambda t}$. The Bochner Laplace transform of the semigroup is $\int_0^\infty e^{-\lambda t} P_t^\Sigma dt = (\lambda - L_\Sigma)^{-1}$, which proves the identity. $\square$

Theorem II.4.5 (Resolvent Fenchel dual and resistance). *Let $K \geq 0$ be self-adjoint with closed form $\mathfrak{q}$, and let $e \in \mathcal{H}$. For $\lambda > 0$,*

$$\langle e, (K + \lambda)^{-1} e \rangle = \sup_{v \in D(\mathfrak{q})} \left\{ 2 \operatorname{Re} \langle e, v \rangle - \mathfrak{q}(v, v) - \lambda \|v\|^2 \right\}. \quad (66)$$

If $P_{\ker K} e = 0$, then monotone convergence gives

$$\lim_{\lambda \downarrow 0} \langle e, (K + \lambda)^{-1} e \rangle = \int_{(0, \infty)} s^{-1} d\nu_e(s) =: \mathcal{R}_K(e) \in [0, \infty]. \quad (67)$$

Finiteness of $\mathcal{R}_K(e)$ is equivalent to boundedness of $v \mapsto \langle e, v \rangle$ on the energy space with norm $\mathfrak{q}(v, v)^{1/2}$; in that case it has a unique weak energy representer modulo $\ker K$.

Proof. For $R_\lambda = (K + \lambda)^{-1}$, completing the square in the form norm of $K + \lambda$ shows that the supremum is attained at $v = R_\lambda e$ and equals $\langle e, R_\lambda e \rangle$. The spectral theorem gives $\langle e, R_\lambda e \rangle = \int (\lambda + s)^{-1} d\nu_e(s)$. Monotone convergence proves (67). The last equivalence is the Riesz theorem on the completion of $D(\mathfrak{q})/\ker K$. $\square$

Theorem II.4.6 (Harmonic defect equals resistance error). *Let $K \geq 0$ be self-adjoint with form $\mathfrak{q}$, and let*

$$\mathcal{E}_K := \overline{D(\mathfrak{q})/\ker K}^{\mathfrak{q}}$$

be its energy space. Suppose $\ell \in \mathcal{E}_K^$, let x solve $\mathfrak{q}(x, \phi) = \ell(\phi)$ weakly modulo $\ker K$, and let $v \in D(\mathfrak{q})$ be a trial state with residual*

$$e_v(\phi) := \mathfrak{q}(v, \phi) - \ell(\phi). \quad (68)$$

Then

$$\|e_v\|_{\mathcal{E}_K^*}^2 = \|[v - x]\|_{\mathcal{E}_K}^2 = \mathfrak{q}(v - x, v - x). \quad (69)$$

If e_v is represented by a vector $e \perp \ker K$ in $\mathcal{H}$, then

$$\|e_v\|_{\mathcal{E}_K^*}^2 = \langle e, K^+ e \rangle = \mathcal{R}_K(e). \quad (70)$$

Thus an approximate committor, elliptic corrector, or resolvent state is controlled by one resistance invariant of its harmonic defect; no $\ker K$ component of the trial state enters the error. When that kernel is a certified gauge or target-null space, the corresponding discarded motion requires no estimate.

Proof. The weak equation and (68) give

$$e_v(\phi) = \mathbf{q}(v - x, \phi).$$

The Riesz map of $\mathcal{E}_K$ is an isometry onto its dual, proving (69). When the residual is represented by e , the spectral theorem identifies its squared energy-dual norm with $\int_{(0,\infty)} s^{-1} d\nu_e(s)$, which is (70); the middle pairing is understood as this spectral quadratic-form value. $\square$

Corollary II.4.7 (Defect-aligned drift bound). *If $e \perp \ker K$ and $\mathcal{R}_K(e) < \infty$, then for every $w \in D(\mathbf{q})$,*

$$|\langle e, w \rangle| \leq \mathcal{R}_K(e)^{1/2} \mathbf{q}(w, w)^{1/2}. \quad (71)$$

Only the component of w paired with the routable defect is charged.

Proof. Use the energy representer from Theorem II.4.5 and apply Cauchy–Schwarz in $\mathcal{E}_K$. $\square$

Proposition II.4.8 (Routable and zero-mode resistance split). *For arbitrary $e \in \mathcal{H}$, write*

$$e = e_0 + e_\perp, \quad e_0 = P_{\ker K} e, \quad e_\perp = P_{(\ker K)^\perp} e.$$

Then

$$\langle e, (K + \lambda)^{-1} e \rangle = \frac{\|e_0\|^2}{\lambda} + \int_{(0,\infty)} \frac{1}{s + \lambda} d\nu_{e_\perp}(s). \quad (72)$$

Consequently the zero-discount resistance is finite precisely when $e_0 = 0$ and $\int_{(0,\infty)} s^{-1} d\nu_{e_\perp}(s) < \infty$. A nonzero target-visible zero mode is an inherited Geom/Caus zero-frequency occurrence, not a residual coordinate. It is removed only when it belongs to a declared gauge or to $\mathcal{N}_\Sigma$; otherwise a separate carrier or price certificate is required.

Proof. The spectral projection of K at 0 contributes $\lambda^{-1} \|e_0\|^2$; the remaining spectral measure is supported on $(0, \infty)$, which gives (72). Monotone convergence yields the finiteness criterion. The source statement follows from Theorems I.3.4 and I.4.5. $\square$

Proposition II.4.9 (Capacity and committor realization). *Let A and B be disjoint quasi-closed sets for a regular or quasi-regular Dirichlet form. The condenser capacity*

$$\text{Cap}(A, B) = \inf \{ \mathbf{q}(v, v) : v \in D(\mathbf{q}), \tilde{v} = 1 \text{ on } A, \tilde{v} = 0 \text{ on } B \}$$

is an analytic Geom certificate. If the minimizer is unique and the associated right process realizes the weak Dirichlet problem, then its quasi-continuous version is the committor $x \mapsto \mathbb{P}_x\{\tau_A < \tau_B\}$ quasi-everywhere. The probabilistic conclusion uses the right-process realization; positive capacity alone does not supply it [3, 7].

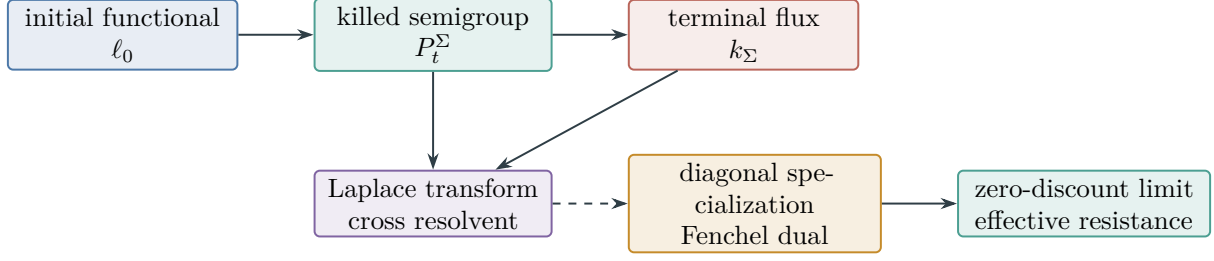


Figure 14: Killed arrival and resistance are related but distinct. Arrival is a cross pairing that requires a flux realization. The Fenchel and resistance identities concern a diagonal self-adjoint specialization.

Theorem II.4.10 (Polar-target capacity certificate). *Let $\mathfrak{q}$ be a symmetric quasi-regular Dirichlet form with associated right process, and let Σ be quasi-closed. Write*

$$\text{Cap}_{\mathfrak{q}}(\Sigma) := \inf\{\mathfrak{q}(v, v) + \|v\|_2^2 : v \in D(\mathfrak{q}), \tilde{v} \geq 1 \text{ q.e. on a neighborhood of } \Sigma\}.$$

If $\text{Cap}_{\mathfrak{q}}(\Sigma) = 0$, then Σ is polar:

$$\mathbb{P}_x\{\tau_{\Sigma} < \infty\} = 0 \quad \text{for quasi-every } x. \quad (73)$$

This is a direct Geom nonreachability certificate for the Markov realization and requires no cascade argument. Positive capacity ends this certificate but supplies no hitting lower bound without a starting-law or nonpolarity statement.

Proof. For a quasi-regular Dirichlet form, sets of zero 1-capacity are properly exceptional up to a capacity-zero enlargement. The associated right process avoids every properly exceptional set for quasi-every starting point, which gives (73) [3, 7]. $\square$

Proposition II.4.11 (Quotient pullback of zero capacity). *Let $q : X \rightarrow Q$ identify a target $\Sigma = q^{-1}(\Sigma_Q)$ quasi-everywhere, and let $\mathfrak{q}_X, \mathfrak{q}_Q$ be symmetric quasi-regular Dirichlet forms. Suppose that for every admissible quotient potential f , with $\tilde{f} \geq 1$ near Σ_Q , there is a lifted potential ψ_f with $\tilde{\psi}_f \geq 1$ near Σ and*

$$(\mathfrak{q}_X)_1(\psi_f, \psi_f) \leq C(\mathfrak{q}_Q)_1(f, f) + \varepsilon(f). \quad (74)$$

If a minimizing family for $\text{Cap}_{\mathfrak{q}_Q}(\Sigma_Q)$ can be chosen with $\varepsilon(f) \rightarrow 0$, then

$$\text{Cap}_{\mathfrak{q}_X}(\Sigma) \leq C \text{Cap}_{\mathfrak{q}_Q}(\Sigma_Q). \quad (75)$$

In particular, zero quotient capacity pulls back to zero capacity on X .

Proof. Every ψ_f is an admissible competitor for the capacity of Σ . Apply (74) along the stated minimizing family and take the liminf. $\square$

II.5 Full-generator response and physical aiming

Definition II.5.1 (Target-active full resolvent). Let $L = -H + A_{\text{Caus}} + B_{\text{Bdry}}$ be a closed generator with a declared resolvent point λ . For closed mathematical input and target-output subspaces $\mathcal{J}$ and $\mathcal{O}_{\Sigma}$, with bounded projections $P_{\mathcal{J}}$ and $P_{\mathcal{O}}$, define

$$\Psi_{\Sigma}(\lambda; L) := \|P_{\mathcal{O}}(\lambda - L)^{-1}P_{\mathcal{J}}\|. \quad (76)$$

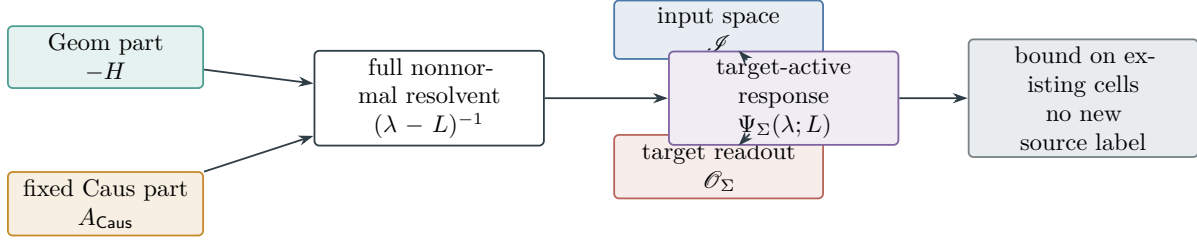


Figure 15: Full nonnormal response retains both Geom and fixed Caus. Input and target compression select a response coefficient; they do not change the operator's source ancestry.

This quantity measures nonnormal amplification visible to the chosen target interface. It is a certificate for the existing Geom–Caus–Bdry generator, not an additional source.

Proposition II.5.2 (Nonnormal source invariance). *Let L_0 be closed on a Banach space and let every K^c be bounded. Put $L_\theta = L_0 + \sum_c \theta_c K^c$ on $D(L_0)$, and suppose $\lambda \in \rho(L_\theta)$ locally in θ . Then the resolvent is norm differentiable and*

$$\partial_{\theta_c}(\lambda - L_\theta)^{-1} = (\lambda - L_\theta)^{-1} K^c (\lambda - L_\theta)^{-1}. \quad (77)$$

Thus every first variation of target-active response is factored through the same source term K^c . Large pseudospectral response can expose a large Geom–Caus interaction, but it creates no new nonnormal source.

Proof. Differentiate $(\lambda - L_\theta)(\lambda - L_\theta)^{-1} = I$ on the common domain and solve for the derivative. Bounded input/output compression preserves the factorization. $\square$

Definition II.5.3 (Target-gradient ray). Let F be a differentiable target shell and assume its mobility gradient

$$G_F := \mathbb{K}(u) DF(u)$$

belongs to $\mathfrak{J}_u$. For a physical current $J \in \mathfrak{J}_u$, define its positive target power and aiming contribution by

$$P_F(J) := (J, G_F)_{u,*}, \quad \text{Align}_F(J) := \frac{(P_F(J)_+)^2}{\|G_F\|_{u,*}^2}, \quad (78)$$

with value zero when $G_F = 0$. By the Cameron–Martin pairing,

$$P_F(J) = \langle DF(u), J \rangle$$

for every $J \in \mathfrak{J}_u$; components of DF in $\ker \mathbb{K}(u)$ annihilate that current space.

Theorem II.5.4 (Projection onto the target ray). *For every $J, G \in \mathfrak{J}_u$,*

$$\|J\|_{u,*}^2 = \frac{((J, G)_{u,*})_+^2}{\|G\|_{u,*}^2} + \text{dist}_{u,*}^2(J, \mathbb{R}_+ G), \quad (79)$$

with the first term zero when $G = 0$. Hence

$$(P_F(J)_+)^2 \leq \|J\|_{u,*}^2 \|G_F\|_{u,*}^2. \quad (80)$$

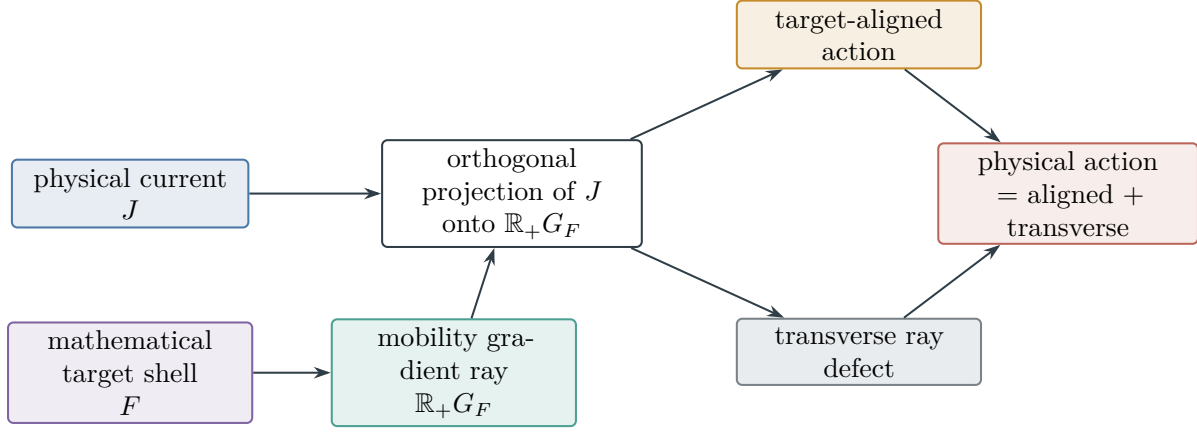


Figure 16: Physical aiming geometry. The target shell is a mathematical test of the already selected current. Projection splits the physical action into target-aligned and transverse expenditure.

Proof. The orthogonal projection of J onto the closed ray $\mathbb{R}_+ G$ has coefficient $((J, G)_{u,*})_+ / \|G\|_{u,*}^2$. Pythagoras gives (79); dropping the nonnegative distance gives (80). $\square$

Proposition II.5.5 (Integrated action–aiming bound). *Let I be a pre-contact interval and assume the two squared norms below are measurable and integrable. Then*

$$\int_I P_F(J(t))_+ dt \leq \left(\int_I \|J(t)\|_{u(t),*}^2 dt \right)^{1/2} \left(\int_I \|G_F(u(t))\|_{u(t),*}^2 dt \right)^{1/2}. \quad (81)$$

For the selected Geom current of Definition I.1.5, the square of the first factor is the physical action and is bounded by the right-hand side of (6).

Proof. Apply (80) and then Cauchy–Schwarz in time. The final assertion is Proposition I.1.6. $\square$

II.6 Degenerate geometry and commutator-generated carriers

Direct resistance controls only the positive spectral subspace of the Geom operator. Fixed Caus transport may nevertheless move a Geom-null direction into a dissipative direction. The correct reduced invariant is then a positive commutator Gramian together with a carrier identity realizing that Gramian along the actual evolution.

Definition II.6.1 (Degenerate Geom–Caus core). Let $\mathcal{H}$ be a complex Hilbert space and

$$L = -S + A, \quad S = S^* \geq 0, \quad A^* = -A,$$

on a common invariant core $\mathcal{D}$. Let $\mathcal{G} \subseteq \ker S$ be a closed A -invariant gauge subspace, and write $P_{\mathcal{G}}^\perp$ for its orthogonal complement. Put $C_0 = S^{1/2}$. When the iterated commutators are defined and closable on $\mathcal{D}$, set

$$C_{k+1} = [A, C_k] \quad \text{on } \mathcal{D}.$$

For fixed K and $c_k > 0$, assume the quadratic form

$$q_K(v) := \sum_{k=0}^K c_k \|C_k v\|^2 \quad (82)$$

is closable. Its positive self-adjoint operator is denoted by Q_K , and its reduced bracket gap is

$$\gamma_K := \inf_{\substack{v \in D(q_K) \cap \mathcal{G}^\perp \\ \|v\|=1}} q_K(v). \quad (83)$$

All domain, closure, and gauge conditions in this definition are part of the certificate; algebraic commutator span alone is insufficient.

Proposition II.6.2 (Bracket resistance invariant). *If $\gamma_K > 0$, then $Q_K \geq \gamma_K I$ on $\mathcal{G}^\perp$, and for every $e \in \mathcal{G}^\perp$,*

$$\mathcal{R}_K^{\text{br}}(e) := \langle e, Q_K^{-1} e \rangle \leq \gamma_K^{-1} \|e\|^2. \quad (84)$$

The quotient kernel

$$\mathfrak{h}_K := \mathcal{G}^\perp \cap \ker Q_K$$

is the structural rank deficit when $\gamma_K = 0$. Every target-visible element of $\mathfrak{h}_K$ retains its Geom–Caus ancestry and remains an existing source occurrence requiring another certificate.

Proof. The first assertion is the definition of the form lower bound in (83); functional calculus gives (84). The kernel is the intersection of the kernels of the closed commutator forms in (82). Source preservation and the target-null residual theorem give the final assertion. $\square$

Definition II.6.3 (Twisted-carrier realization certificate). A bracket Gramian Q_K has a twisted-carrier realization when there is a bounded self-adjoint operator M on $\mathcal{G}^\perp$, constants $0 < m_- \leq m_+ < \infty$ and $c_M > 0$, and a common evolution core on which

$$m_- \|v\|^2 \leq \langle Mv, v \rangle \leq m_+ \|v\|^2, \quad (85)$$

$$2 \operatorname{Re} \langle Mv, Lv \rangle \leq -c_M q_K(v). \quad (86)$$

For a forced equation, the positive part of $2 \operatorname{Re} \langle Mu, f_{\text{Bdry}} \rangle dt$ is included in the input measure R_u^+ .

Theorem II.6.4 (Hypocoercive carrier produces physical expenditure). *Assume Definition II.6.3, and let a legal orbit satisfy $\dot{u} = Lu + f_{\text{Bdry}}$ on the declared common core, with the chain rule for $\langle Mu, u \rangle$. Then*

$$\mathcal{K}_u(t) := \langle Mu(t), u(t) \rangle, \quad \mathfrak{D}_u(dt) := c_M q_K(u(t)) dt, \quad \Phi_u^+ = 0$$

satisfy the carrier balance of Definition II.2.4. Hence

$$\mu_u([0, T_*)) \leq \langle Mu_0, u_0 \rangle + R_u^+([0, T_*)) \leq m_+ \|u_0\|^2 + R_u^+([0, T_*)) \quad (87)$$

after subtracting any known lower carrier level. If $\gamma_K > 0$, this expenditure controls every nongauge direction in $\mathcal{G}^\perp$. The commutator calculation is therefore a structural compiler for a physical budget in a partially dissipative system [10].

Proof. Differentiate $\langle Mu, u \rangle$ and use (86); the positive forcing pairing gives the declared input measure. Apply Theorem II.2.5. When $\gamma_K > 0$, $q_K(u) \geq \gamma_K \|P_{\mathcal{G}}^\perp u\|^2$. $\square$

Definition II.6.5 (Positive-commutator carrier). Let the evolution be $\dot{u} = -iHu$, with $H = H^*$, and let $P = \mathbf{1}_\Delta(H)$ be a reducing spectral projection. A self-adjoint conjugate operator C supplies a positive-commutator carrier when the form

$$B := i[H, C]$$

is well defined and nonnegative on a common invariant core in $P\mathcal{H}$, and the expectation $\langle u(t), Cu(t) \rangle$ has a finite upper bound C_+ along the orbit. A strict Mourre certificate has $PBP \geq \theta P$ modulo the declared gauge for some $\theta > 0$.

Theorem II.6.6 (Conservative carrier produces physical expenditure). *Under Definition II.6.5, let $u_0 \in P\mathcal{H}$ and assume the commutator chain rule. Then*

$$\frac{d}{dt} \langle u(t), Cu(t) \rangle = \langle u(t), Bu(t) \rangle. \quad (88)$$

Consequently

$$\mu_u(Q) := \int_Q \langle u(t), Bu(t) \rangle dt, \quad \mu_u([0, T_*)) \leq C_+ - \langle u_0, Cu_0 \rangle. \quad (89)$$

When $PBP \geq \theta P$, this budget controls $\theta \int_0^{T_*} \|Pu(t)\|^2 dt$. Thus a virial, Morawetz, or Mourre identity contributes to the same physical-budget calculus whenever its carrier variation is finite [4].

Proof. Differentiate the expectation on the common core to obtain (88). Nonnegativity of B makes the right side a measure density. Integrate and use the upper carrier bound. The strict estimate follows from $PBP \geq \theta P$. $\square$

Corollary II.6.7 (Complete routing alternatives for a zero mode). *Let e_0 be the zero mode in Proposition II.4.8. Direct resistance controls $e_\perp$. A target-visible e_0 may be controlled by a twisted-carrier certificate or a positive-commutator carrier, with its price charged to the same μ_u . If neither certificate is supplied, the corresponding Geom–Caus source profile remains unevaluated; it does not move to J_{res} .*

II.7 Compactness and rigidity as a price compiler

Compactness is not an additional continuum source and is not part of the definition of the residual. Its role is quantitative: it converts a qualitative zero-price rigidity statement into a uniform positive price on a closed class of target-approaching windows. The topology used in this conversion is local to the certificate.

Definition II.7.1 (Typed compactness–rigidity certificate). Fix a physical expenditure ceiling B_{phys} . A typed compactness–rigidity certificate is a quadruple

$$(X_{\text{cert}}, K_{B_{\text{phys}}}, B_\Sigma, \mathfrak{d})$$

with the following properties.

- (K1) X_{cert} is the explicitly declared Hausdorff topological space in which normalized windows are compared.
- (K2) $K_{B_{\text{phys}}} \subset X_{\text{cert}}$ is compact and contains the closure of every normalized window selected under the ceiling B_{phys} .
- (K3) $B_\Sigma \subset X_{\text{cert}}$ is closed, and $K_{B_{\text{phys}}} \cap B_\Sigma \neq \emptyset$. Membership in B_Σ is the target-approaching condition retained by the normalization.
- (K4) $\mathfrak{d} : K_{B_{\text{phys}}} \rightarrow [0, \infty)$ is lower semicontinuous and is dominated by a localizable physical expenditure as specified in Definition II.7.2.
- (K5) The zero-price rigidity statement

$$\{w \in K_{B_{\text{phys}}} : \mathfrak{d}(w) = 0\} \cap B_\Sigma = \emptyset$$

holds.

The certificate is *typed*: changing the topology, normalization, bad class, or physical functional changes the certificate and requires those clauses to be checked again.

Definition II.7.2 (Physical price certificate). Let Q be a measurable pre-contact event for a legal orbit u , and let $\mathcal{N}_{Qu} \in X_{\text{cert}}$ be its normalized window. A physical price certificate relative to the orbit expenditure measure μ_u consists of a constant $c_{\text{pay}} > 0$ for which

$$\mu_u(Q) \geq c_{\text{pay}} \mathfrak{d}(\mathcal{N}_{Qu}) \quad (90)$$

for every event in the selected family. The normalization may use **Abs**- or **Lift**-operations, but (90) must compare the normalized functional with the measure in the original physical variables. Scale invariance or a change of clock alone supplies no such comparison.

Lemma II.7.3 (Compactness–rigidity price gap). *Let $(X_{\text{cert}}, K_{B_{\text{phys}}}, B_{\Sigma}, \mathfrak{d})$ satisfy Definition II.7.1. Then*

$$\delta_{\text{rig}}(B_{\text{phys}}) := \min_{w \in K_{B_{\text{phys}}} \cap B_{\Sigma}} \mathfrak{d}(w) > 0. \quad (91)$$

Proof. The intersection $K_{B_{\text{phys}}} \cap B_{\Sigma}$ is nonempty and compact. Lower semicontinuity therefore gives a minimizer w_* . If its value were zero, then w_* would belong to the prohibited intersection in clause (K5). Hence the minimum is positive. $\square$

Theorem II.7.4 (Uniform physical price). *Assume a typed compactness–rigidity certificate and a physical price certificate. Every selected event Q whose normalized window belongs to B_{Σ} satisfies*

$$\mu_u(Q) \geq \delta_{\text{phys}}(B_{\text{phys}}), \quad \delta_{\text{phys}}(B_{\text{phys}}) := c_{\text{pay}} \delta_{\text{rig}}(B_{\text{phys}}) > 0. \quad (92)$$

Proof. By selection, $\mathcal{N}_{Qu} \in K_{B_{\text{phys}}} \cap B_{\Sigma}$. Apply (91) in (90). $\square$

Theorem II.7.5 (Borderline passivity price compiler). *Assume $\beta_*^{\text{red}} = 1$ in Definition II.2.6, and define the normalized carrier deficit*

$$\Delta_{\text{car}}(V) := D(V) - \Phi_+(V) \geq 0. \quad (93)$$

Let $K_{B_{\text{phys}}}$ be a compact normalized profile class, $B_{\Sigma} \subseteq K_{B_{\text{phys}}}$ a nonempty closed target-bad class, and assume Δ_{car} is lower semicontinuous with

$$\{V \in K_{B_{\text{phys}}} : \Delta_{\text{car}}(V) = 0\} \cap B_{\Sigma} = \emptyset. \quad (94)$$

If physical-clock pullback gives

$$\mu_u(Q) \geq c_{\text{clock}} \Delta_{\text{car}}(\mathcal{N}_{Qu}), \quad c_{\text{clock}} > 0, \quad (95)$$

then every selected target-bad event has price

$$\mu_u(Q) \geq c_{\text{clock}} \min_{V \in K_{B_{\text{phys}}} \cap B_{\Sigma}} \Delta_{\text{car}}(V) > 0. \quad (96)$$

Thus the critical value $\beta_^{\text{red}} = 1$ is closed by an equality–rigidity check on the reduced saturation class; no strict global flux estimate is required.*

Proof. Apply Lemma II.7.3 with $\mathfrak{d} = \Delta_{\text{car}}$, then use (95). $\square$

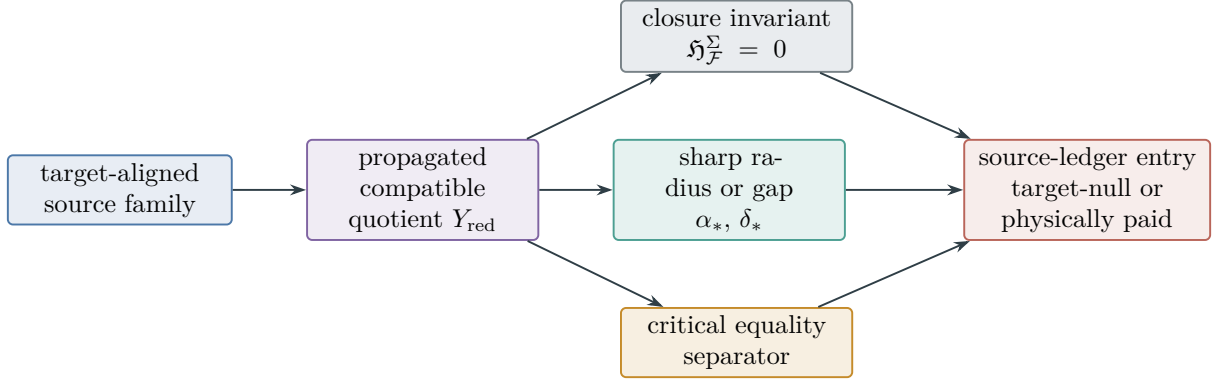


Figure 17: Estimate-to-invariant compiler. Once certified null directions are propagated out, the remaining analytic problem reduces to a target-closure test, a sharp domination radius or physical-price gap, or a critical equality separator. Compactness–rigidity is one method for proving the gap, not an additional structural mechanism.

Corollary II.7.6 (Separator closure of the borderline branch). *Suppose every zero-deficit profile in $K_{B_{\text{phys}}}$ satisfies the equality identities of Proposition II.1.7, and the resulting target-null equality class is disjoint from B_{Σ} . Then (94) holds, so Theorem II.7.5 supplies the uniform event price.*

Proposition II.7.7 (A standard local compactness compiler). *Let X_0, X, X_1 be Banach spaces with $X_0 \Subset X \hookrightarrow X_1$, the first embedding compact and the second continuous. If $1 \leq p < \infty$, $1 < q \leq \infty$, and a family of windows is bounded in*

$$L^p(0, 1; X_0) \cap W^{1,q}(0, 1; X_1),$$

then it is relatively compact in $L^p(0, 1; X)$. Consequently, its closure can serve as $K_{B_{\text{phys}}}$ when the target condition is closed and the price is lower semicontinuous in this topology.

Proof. The relative compactness is the Aubin–Lions lemma in the stated range [1, 6, 9]. The final assertion is the definition of closure together with the remaining explicitly stated hypotheses. $\square$

Remark II.7.8 (Quotients do not supply missing compactness). If a compact group acts continuously on a compact normalized set, its quotient is compact. For a noncompact symmetry group, fixing a gauge only defines a slice; precompactness of that slice and closedness of the gauge must be proved. In either case the quotient operation has **Abs**-ancestry and does not alter the source of the limiting physical record.

Remark II.7.9 (What failure of compactness means). If clause (K2), (K3), (K4), or (K5) is not known, Theorem II.7.4 is unavailable. The relevant **Geom/Caus/Abs/Lift/Bdry** source cell is then unevaluated by this compiler. No target-visible element is transferred to J_{res} .

II.8 Constraint tails as finite structural obstructions

Nonlocal constraints often enter a local evolution through an elliptic multiplier, potential, or pressure. A direct estimate of the whole nonlocal field obscures the smaller structural question: which part is fixed by the local source, which part is a gauge, which part decays from the exterior, and which finite-dimensional harmonic mode remains visible to the target? The following interface makes that separation explicit.

Definition II.8.1 (Elliptic constraint–tail interface). Let $U_0 \Subset U$ be open sets, let

$$\mathcal{C}p = S(V) \quad \text{in } U$$

be a linear elliptic constraint in a declared weak sense, and let $\mathcal{G}_{U_0}$ be its gauge space on U_0 . A constraint–tail interface consists of:

- (T1) a local solution operator $\mathcal{L}_U$ satisfying $\mathcal{C}\mathcal{L}_U S(V) = S(V)$ on U ;
- (T2) a decomposition on U_0 ,

$$p|_{U_0} = p_{\text{loc}} + h, \quad p_{\text{loc}} := \mathcal{L}_U S(V)|_{U_0}, \quad \mathcal{C}h = 0 \text{ in } U_0; \quad (97)$$

- (T3) a Banach space $\mathcal{H}(U_0)$ of homogeneous solutions with a continuous splitting

$$\mathcal{H}(U_0) = \mathcal{H}_{\text{low}} \oplus \mathcal{H}_{\text{dec}}, \quad (98)$$

where $\mathcal{H}_{\text{low}}$ is finite dimensional;

- (T4) a dyadic exterior expansion $P_{\text{dec}}h = \sum_{k \geq 0} h_k$ convergent in $\mathcal{H}_{\text{dec}}$, with

$$\mathfrak{s}_{\text{tail}}(h) := \sum_{k \geq 0} \|h_k\|_{\mathcal{H}_{\text{dec}}} < \infty; \quad (99)$$

- (T5) a linear gauge realization $\Gamma : \mathcal{G}_{U_0} \rightarrow \mathcal{H}_{\text{low}}$ with closed range.

The residual low-mode obstruction is the cokernel class

$$\mathfrak{o}_{\text{tail}}(p) := [P_{\text{low}}h] \in \mathcal{H}_{\text{low}} / \text{Ran } \Gamma. \quad (100)$$

The local term retains the ancestry of $S(V)$; gauge removal appends **Abs**, and the exterior expansion and its limiting trace append **Bdry**.

Theorem II.8.2 (Constraint–tail structural normal form). *Every field in an elliptic constraint–tail interface has, on U_0 , the decomposition*

$$p = p_{\text{loc}} + \Gamma g + h_{\text{dec}} + h_{\text{obs}}, \quad (101)$$

where $g \in \mathcal{G}_{U_0}$, $h_{\text{dec}} \in \mathcal{H}_{\text{dec}}$, and h_{obs} is any representative of the class $\mathfrak{o}_{\text{tail}}(p)$. The first three terms are closed by, respectively, the local source law, a propagated quotient, and the summability invariant (99). Consequently:

- (C1) if $\mathfrak{o}_{\text{tail}}(p) = 0$, the entire constraint field closes using only its inherited source, **Abs**, and **Bdry**;
- (C2) if $\mathfrak{o}_{\text{tail}}(p) \neq 0$ and its target projection vanishes, its remaining class lies in $\mathcal{N}_{\Sigma}$;
- (C3) if $\mathfrak{o}_{\text{tail}}(p) \neq 0$ and is target-visible, it is a finite-dimensional **Bdry/Abs**-ancestry occurrence requiring a carrier or price certificate. It is not a new residual source.

Thus control of a nonlocal constraint reduces to one local source estimate, one summability check, and one finite-dimensional target-projected cokernel.

Proof. Apply the continuous projections in (98) to h . Closed range and finite dimensionality give $\mathcal{H}_{\text{low}} = \text{Ran } \Gamma \oplus H_{\text{obs}}$ for a finite-dimensional complement H_{obs} . This gives (101); changing the complement changes h_{obs} only by a gauge term and hence preserves the cokernel class. Absolute convergence in (99) closes the decaying series. The three routing conclusions follow from quotient propagation, Theorem I.3.4, and the definition of $\mathcal{N}_{\Sigma}$. $\square$

Proposition II.8.3 (Constraint tails enter the physical budget only through a carrier law). *Suppose a constraint tail contributes a signed term Ψ_{tail} to a localized carrier balance. It is charged to the physical expenditure measure only if the equation supplies a measure inequality of one of the forms*

$$(\Psi_{\text{tail}})_+ \leq \Phi_u^+, \quad \text{or} \quad (\Psi_{\text{tail}})_+ \leq R_u^+, \quad (102)$$

with the right side appearing in (53). Tail summability alone proves analytic closure but supplies no physical price and no budget bound.

Proof. The carrier compiler uses only the measure terms displayed in (53). Either inequality in (102) places the positive tail contribution in that balance. Without such a comparison, (99) is a norm statement on a homogeneous field and has no implication for μ_u . $\square$

II.9 Covariant entropy cocycles and physical production

An entropy enters the reachability argument through the physical production that it controls along one orbit. Local entropy formulas in changing blow-up charts must therefore be joined by an explicit cocycle law. Independent windowwise monotonicity does not telescope. The formulation below extracts the equation-independent contracts from the covariant entropy construction in [2].

Definition II.9.1 (Physical entropy cocycle). Let $I_j = [a_j, b_j]$, $j \geq 1$, be ordered normalized windows obtained from pairwise disjoint physical events Q_j . A physical entropy cocycle on this family consists of

- (E1) scalar primitives $\mathcal{M}_j : I_j \rightarrow \mathbb{R}$;
 - (E2) nonnegative Radon production measures $\mathfrak{D}_j$ and nonnegative error measures r_j on I_j ;
 - (E3) reset errors $e_j \geq 0$ and a constant $c_{\mathcal{M}} > 0$;
 - (E4) constants $M_+ < \infty$ and $M_- > -\infty$;
- such that, for $a_j \leq s < t \leq b_j$,

$$\mathcal{M}_j(t) + c_{\mathcal{M}} \mathfrak{D}_j((s, t]) \leq \mathcal{M}_j(s) + r_j((s, t]), \quad (103)$$

$$\mathcal{M}_{j+1}(a_{j+1}) \leq \mathcal{M}_j(b_j) + e_j, \quad (104)$$

$$\mathcal{M}_1(a_1) \leq M_+, \quad \mathcal{M}_j(t) \geq M_-. \quad (105)$$

The measures $\mathfrak{D}_j$ are coordinate expressions of one physical production law on the events Q_j ; chart Jacobians and time changes are included in that law. The half-open convention in (103) prevents endpoint atoms from being counted twice.

Theorem II.9.2 (Entropy-cocycle gluing). *For every $N \geq 1$, a physical entropy cocycle satisfies*

$$c_{\mathcal{M}} \sum_{j=1}^N \mathfrak{D}_j(I_j) \leq M_+ - M_- + \sum_{j=1}^N r_j(I_j) + \sum_{j=1}^{N-1} e_j. \quad (106)$$

If the two error series are finite, then $\sum_j \mathfrak{D}_j(I_j) < \infty$. If in addition every selected bad window satisfies $\mathfrak{D}_j(I_j) \geq \delta > 0$, their number is at most

$$\frac{M_+ - M_- + \sum_j r_j(I_j) + \sum_j e_j}{c_{\mathcal{M}} \delta}. \quad (107)$$

Proof. Apply (103) on every whole window. Summing gives

$$c_{\mathcal{M}} \sum_{j=1}^N \mathfrak{D}_j(I_j) \leq \sum_{j=1}^N (\mathcal{M}_j(a_j) - \mathcal{M}_j(b_j)) + \sum_{j=1}^N r_j(I_j).$$

The reset inequalities telescope the first sum from above:

$$\sum_{j=1}^N (\mathcal{M}_j(a_j) - \mathcal{M}_j(b_j)) \leq \mathcal{M}_1(a_1) - \mathcal{M}_N(b_N) + \sum_{j=1}^{N-1} e_j.$$

Use (105). This proves (106). Monotone convergence gives the countable statement, and $N\delta \leq \sum_{j=1}^N \mathfrak{D}_j(I_j)$ gives (107). $\square$

Certificate II.9.3 (Perelman-type covariant compiler). A Perelman-type construction is a valid instance of Definition II.9.1 only after the following facts are proved for the PDE and the chosen window atlas.

- (P1) Each local primitive is a quotient-invariant scalar, typically a localized energy or variance plus an adjoint entropy and explicit pressure, drift, modulation, and boundary corrections.
- (P2) Differentiation along the *fixed* PDE gives (103); all unsigned remainders are placed in the displayed error measure, not suppressed formally.
- (P3) The production contains a nonnegative lower-semicontinuous physical floor $\mathfrak{d}$ that can be used in Definition II.7.1.
- (P4) The primitives have the lower bound in (105). Its sign must match the orientation of (103).
- (P5) Transport along the same physical orbit, chart pullback, and any change of a finite-horizon adjoint datum satisfy (104), with summable reset errors.
- (P6) Production either defines the expenditure measure μ_u , as in Corollary II.9.6, or is dominated by the declared μ_u through an explicit change-of-variables inequality of the form (90).

These clauses isolate the production-and-cocycle mechanism of Perelman's entropy formula [8]; existence of such a scalar is an equation-specific certificate.

Proposition II.9.4 (Weighted zero-production rigidity). *Let $D \subset \mathbb{R}^d$ be connected and open, let $v \in L^2(I; W_{\text{loc}}^{1,2}(D))$, and let $m \geq m_* > 0$ almost everywhere on $D \times I$. If a nonnegative production functional obeys*

$$\mathfrak{d}(w) \geq c_0 \int_I \int_D m(x, t) |\nabla v(x, t)|^2 dx dt \quad (108)$$

with $c_0 > 0$, then $\mathfrak{d}(w) = 0$ implies that $v(\cdot, t)$ is spatially constant on D for almost every t . Consequently, any closed bad class carrying a strictly positive mean-subtracted oscillation is disjoint from $\{\mathfrak{d} = 0\}$.

Proof. Nonnegativity and (108) imply $\nabla v = 0$ almost everywhere. A Sobolev function with zero weak gradient on a connected domain is almost everywhere constant. Its mean-subtracted oscillation is therefore zero, which excludes membership in the stated bad class. $\square$

Proposition II.9.5 (Physical-clock comparison). *Let μ be a physical measure on an event and suppose a normalized production measure has density $w \geq 0$ with respect to it: $d\nu = w d\mu$. On a measurable set Q ,*

$$w \leq C, \quad 0 < C < \infty, \quad \text{a.e. on } Q \implies \mu(Q) \geq C^{-1} \nu(Q), \quad (109)$$

$$w \geq c > 0 \quad \text{a.e. on } Q \implies \mu(Q) \leq c^{-1} \nu(Q). \quad (110)$$

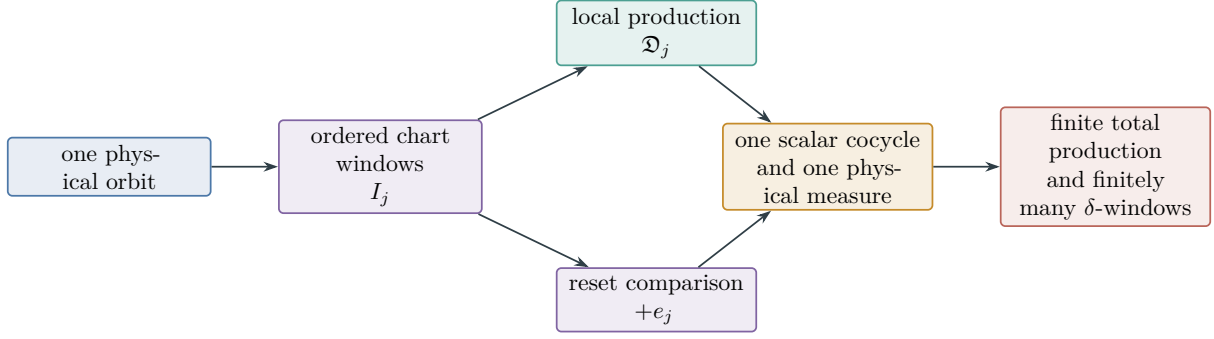


Figure 18: Covariant entropy-cocycle accounting. Local production estimates become global only after chart and adjoint-reset comparisons identify them along the same physical orbit. The output is a finite physical production measure that can serve as the orbit expenditure μ_u .

Thus a lower bound for normalized production yields a lower physical price only from an upper bound such as (109); the opposite density bound has the opposite implication.

Proof. Integrate the pointwise bounds in $\nu(Q) = \int_Q w \, d\mu$. $\square$

Corollary II.9.6 (Entropy as the physical expenditure measure). *Under Definition II.9.1, assume the total error in (106) is finite. On the atomic σ -algebra generated by the disjoint events Q_j , define*

$$\nu_u \left(\bigsqcup_{j \in S} Q_j \right) := \sum_{j \in S} \mathfrak{D}_j(I_j), \quad S \subseteq \mathbb{N}.$$

Then ν_u is a finite, countably additive physical measure on that event family. It may therefore be chosen as μ_u in Definition I.1.3. If other expenditures must also be charged, the measure may be enlarged only after a carrier or global balance proves a finite bound for the combined sum; that combined measure is fixed before the single ceiling B_{phys} is set.

Proof. Countable additivity follows directly from the nonnegative series defining ν_u ; finiteness is Theorem II.9.2. $\square$

Remark II.9.7 (Equality models are certificate-local). A Gaussian may be the equality state of an adjoint Fisher-information term, while a constant field is the equality state of a weighted Dirichlet floor. These statements concern different nonnegative terms. A valid rigidity argument uses the equality set of the particular floor that is both compactly stable and linked to target badness; it does not merge the two equality models.

Part III

Structural Reachability and Regularity

III.1 Singular targets and cascade compilers

The algebra and normalized shell identity first decide whether target-aligned structure exists. Its absence excludes contact directly. On the surviving aligned branch, exclusion requires two

problem-specific facts: target contact must generate an infinite family of disjoint pre-contact events, and every selected persistent source cell must carry lower event prices with divergent total mass in one finite physical measure. Shell accountability, cascade extraction, and physical pricing are separate implications.

Definition III.1.1 (Singular target). Let X_{reg} be the declared regular state class and let $\mathcal{G}$ be a Hausdorff state-or-germ space containing its admissible limits. The singular target $\Sigma_{\text{sing}} \subset \mathcal{G}$ is the set of nonremovable germs at which the declared regular solution cannot be continued. For a maximal regular orbit $u : [0, T_{\text{max}}) \rightarrow X_{\text{reg}}$, contact means that there are $t_n \uparrow T_{\text{max}}$ for which $u(t_n)$ converges in the target topology to an element of Σ_{sing} . The target is assumed closed in that topology. This definition makes no assertion that finite-time noncontinuation actually implies contact; that implication is the continuation hypothesis in Corollary III.2.5.

Definition III.1.2 (Singularity-formation support). For a legal orbit u with a normalized target-shell presentation, let $\mathcal{D}_\Sigma(u)$ be a family of pre-contact shell records

$$R_r = (I_r, \Phi_r, (Q_{c,r}(u))_{c \in \mathfrak{C}})$$

attached to that same orbit. The family is *shell-complete* when contact implies that it contains such a record for every sufficiently small r . Every record supplied by contact then satisfies

$$\text{Prog}_\Sigma(R_r) = \sum_{c \in \mathfrak{C}} Q_{c,r}(u) = 1. \quad (111)$$

The singularity-formation support is

$$\mathfrak{S}_\Sigma(u) := \bigcup_{R \in \mathcal{D}_\Sigma(u)} \text{Align}_\Sigma(R) \subseteq \mathfrak{C}. \quad (112)$$

It answers the target-relative question: which inherited source currents actually have positive pairing with the covectors descending toward the singular set? It does not ask which terms of the PDE are large in a target-independent norm.

Proposition III.1.3 (A singularity requires aligned structure). *If u is generated by the compiled continuum algebra and $\mathcal{D}_\Sigma(u)$ is shell-complete, then*

$$u \text{ contacts } \Sigma \implies \mathfrak{S}_\Sigma(u) \neq \emptyset. \quad (113)$$

Consequently $\mathfrak{S}_\Sigma(u) = \emptyset$ implies target avoidance. This implication is independent of the size or compactness of the full orbit.

Proof. Contact supplies normalized records satisfying (111). By Theorems I.5.2 and I.5.12, each has a nonempty aligned support, contained in the union (112). The contrapositive gives target avoidance. $\square$

Theorem III.1.4 (Canonical disjoint shell cascade). *Let a compiled legal orbit contact Σ at a finite time or converge to a target germ at its maximal time, and let (U_r, Φ_r) be a target-shell presentation. There are scales $r_j \downarrow 0$ and pairwise disjoint, chronologically ordered, strictly pre-contact intervals $I_j = [a_j, b_j]$ such that*

$$u(a_j) \notin U_{2r_j}(\Sigma), \quad u(b_j) \in U_{r_j}(\Sigma), \quad \text{Prog}_\Sigma(I_j, \Phi_{r_j}) = 1. \quad (114)$$

Every interval therefore has a nonempty aligned support and a source contribution at least $1/5$. No compactness, price gap, or contact margin is used in this extraction.

Proof. Set $b_0 = 0$. Since every strictly pre-contact state is outside $\Sigma = \bigcap_{r>0} U_{2r}(\Sigma)$, after b_{j-1} choose a regular time $a_j > b_{j-1}$ and then a scale $r_j < \min\{r_{j-1}/4, 1/j\}$ for which $u(a_j) \notin U_{2r_j}(\Sigma)$. Contact supplies a later strictly pre-contact time at which the orbit belongs to $U_{r_j}(\Sigma)$; let b_j be the first such time after a_j . This inductively gives $b_{j-1} < a_j < b_j$, hence pairwise disjoint intervals. Continuity in the shell topology supplies the stated endpoint values (or their harmless one-sided limiting version, after shortening the interval). The endpoint normalization gives unit progress. Apply Theorem I.5.2 on each interval to get the final assertions. $\square$

Definition III.1.5 (Renormalization-scale interface). Let z_* be a putative contact point or germ. A renormalization-scale interface consists of scales $r_j \downarrow 0$, physical pre-contact events $Q_j = Q(z_*, r_j)$, and legal normalizations

$$V_j = \mathcal{S}_{z_*, r_j} u \in X_{\text{cert}},$$

where every $\mathcal{S}_{z,r}$ has its declared Abs/Lift ancestry. It has:

- (R1) *geometric separation* if, after passing to a subsequence, $r_{j+1} \leq \vartheta r_j$ for some $0 < \vartheta < 1$ and the events Q_j are pairwise disjoint;
- (R2) *target persistence* if every V_j lies in one closed normalized bad class B_Σ ;
- (R3) a *physical scaling law* if there are $c_{\text{sc}} > 0$, $\kappa \in \mathbb{R}$, and a nonnegative normalized functional $\mathfrak{d}_{\text{sc}}$ such that

$$\mu_u(Q_j) \geq c_{\text{sc}} r_j^\kappa \mathfrak{d}_{\text{sc}}(V_j). \quad (115)$$

The exponent κ is computed from the physical measure in the original variables, not from a freely chosen normalized clock.

Theorem III.1.6 (Scale-cascade price compiler). *Assume a renormalization-scale interface has geometric separation and target persistence, and suppose*

$$\delta_{\text{sc}} := \inf_{V \in B_\Sigma} \mathfrak{d}_{\text{sc}}(V) > 0. \quad (116)$$

If the physical scale weights have divergent total mass,

$$\sum_{j=1}^{\infty} r_j^\kappa = \infty, \quad (117)$$

then

$$B_{\text{phys}} \geq \mu_u \left(\bigsqcup_{j=1}^{\infty} Q_j \right) = \sum_{j=1}^{\infty} \mu_u(Q_j) \geq c_{\text{sc}} \delta_{\text{sc}} \sum_{j=1}^{\infty} r_j^\kappa = \infty, \quad (118)$$

contradicting a finite physical budget. The stronger condition $a_ := \inf_j r_j^\kappa > 0$ gives the uniform event price $\mu_u(Q_j) \geq c_{\text{sc}} a_* \delta_{\text{sc}} > 0$. After restricting to $r_j \leq 1$, the divergent-series condition is automatic when $\kappa \leq 0$, and the scale-critical case $\kappa = 0$ transfers the normalized gap without loss. When $\kappa > 0$, the exact criterion is the divergence of the scale-weight series; a normalized gap alone supplies no such conclusion.*

Proof. Insert (116) into (115) and sum over the pairwise disjoint events. This gives (118); countable additivity and the finite physical budget give the contradiction. If $0 < r_j \leq 1$, then $r_j^\kappa \geq 1$ for $\kappa \leq 0$. For $\kappa > 0$, the individual weights tend to zero, so their summability must be decided from the actual scale sequence. $\square$

Proposition III.1.7 (Structural data required from a blow-up argument). *Within the scale interface, the analytic blow-up argument is used only to establish the following invariant data:*

$$\left(\vartheta, \overline{\{V_j\}}, \mathfrak{H}_{\{V_j\}}^\Sigma, \kappa, \delta_{\text{sc}} \right). \quad (119)$$

Here $\vartheta < 1$ certifies disjoint scale extraction, the closure is taken in the stated certificate topology, $\mathfrak{H}_{\{V_j\}}^\Sigma$ records whether the cascade remains target-visible, κ is the physical scaling exponent, and δ_{sc} is the price gap. Once this tuple has been verified, no estimate on a source direction already removed by a compatible quotient is used in the exclusion proof.

Definition III.1.8 (Normalized cascade refinement). Fix a legal physical presentation generated by the compiled continuum algebra and a closed target-bad class $B_\Sigma \subset X_{\text{cert}}$. A normalized cascade refinement consists of the canonical intervals supplied by Theorem III.1.4, together with

- (C1) pairwise disjoint measurable pre-contact events $Q_1, Q_2, \dots$ in the domain of the same physical expenditure measure μ_u ;
- (C2) legal normalizations $\mathcal{N}_{Q_j}$, using only the declared Abs/Lift operations, such that $w_j := \mathcal{N}_{Q_j}u \in B_\Sigma$;
- (C3) for every j , a normalized shell-crossing record R_j carried by w_j for which

$$\text{Prog}_\Sigma(R_j) = 1, \quad \max_{c \in \mathfrak{C}} (\text{Prog}_\Sigma^c(R_j))_+ \geq \frac{1}{5}. \quad (120)$$

By Theorem I.4.5 and the applicable prospective shell-accountability theorem, every occurrence in clause (C3) has a nonempty aligned source support $A_j := \text{Align}_\Sigma(R_j) \subseteq \mathfrak{C}$. This refinement adds only equation-specific normalizations to the already proved shell cascade. It does not assert a positive price.

Proposition III.1.9 (Persistent channel extraction). *For a cascade supplied by Definition III.1.8, some nonempty support $A \subseteq \mathfrak{C}$ occurs for infinitely many indices. In particular, at least one of the five elementary source labels belongs to the ancestry of infinitely many target-approaching windows.*

Proof. There are only $2^5 - 1$ nonempty supports. The infinite pigeonhole principle gives an infinite constant-support subsequence. Any member of that support has the second asserted property. $\square$

Definition III.1.10 (Target-aligned source ledger). Fix a legal orbit u and a normalized cascade family $(Q_j, R_j)_{j \geq 1}$. For every nonempty $A \subseteq \mathfrak{C}$, let $\mathcal{F}_A(u)$ be the family of source-current occurrences extracted from those R_j for which $\text{Align}_\Sigma(R_j) = A$. The source ledger is *complete* when each such A has one of the following entries:

(L1) a *null entry*,

$$\mathfrak{H}_{\mathcal{F}_A(u)}^\Sigma = 0; \quad (121)$$

(L2) a *paid entry*: writing $I_A := \{j : \text{Align}_\Sigma(R_j) = A\}$, there is a nonnegative price schedule $(\delta_{u,A,j})_{j \in I_A}$ such that

$$\mu_u(Q_j) \geq \delta_{u,A,j} \quad (j \in I_A), \quad |I_A| = \infty \implies \sum_{j \in I_A} \delta_{u,A,j} = \infty. \quad (122)$$

A null entry may be proved by quotient, symmetry, or conservation. A paid entry may be proved directly or compiled from a carrier radius, resistance, compactness–rigidity gap, equality separator, scale law, or entropy production. The entry records only the resulting invariant, not the analytic route used to establish it. A uniform lower price is the special schedule $\delta_{u,A,j} \equiv \delta_{u,A} > 0$; scale-dependent schedules are allowed whenever their sum diverges. Since $|\mathfrak{C}| = 5$, the ledger has at most $2^5 - 1 = 31$ nonempty rows even when the cascade contains infinitely many scales.

Theorem III.1.11 (Physical expenditure exclusion). *Let μ be a nonnegative countably additive measure with $\mu(\Omega) \leq B_{\text{phys}} < \infty$. Let $Q_j \subseteq \Omega$ be pairwise disjoint measurable sets and let $\delta_j \geq 0$. If*

$$\mu(Q_j) \geq \delta_j \quad \text{for every } j, \quad \sum_{j=1}^{\infty} \delta_j = \infty, \quad (123)$$

then such a family cannot exist. The uniform-price criterion $\delta_j \geq \delta > 0$ is a special case.

Proof. For every N , finite additivity and monotonicity give

$$B_{\text{phys}} \geq \mu\left(\bigsqcup_{j=1}^N Q_j\right) = \sum_{j=1}^N \mu(Q_j) \geq \sum_{j=1}^N \delta_j.$$

The partial sums on the right are unbounded by (123), contradicting the fixed upper bound B_{phys} . $\square$

Corollary III.1.12 (Shell progress implies expenditure exclusion). *Suppose contact supplies normalized shell records R_j on pairwise disjoint pre-contact events Q_j , and suppose a physical certificate proves*

$$\text{Prog}_{\Sigma}(R_j) = 1 \quad \implies \quad \mu_u(Q_j) \geq \delta_j \geq 0. \quad (124)$$

If $\sum_j \delta_j = \infty$, then u cannot contact Σ .

Proof. Contact supplies the unit identities by Theorem I.5.2. The price implication then gives (123), and Theorem III.1.11 gives the contradiction. $\square$

Theorem III.1.13 (Finite structural-ledger exclusion). *Let u have the canonical contact cascade of Theorem III.1.4, and suppose every persistent target-aligned support has been structurally reduced either to zero in the target quotient or to a divergent physical-price schedule. Then u does not contact Σ .*

Proof. Assume contact. By Proposition III.1.9, there is a nonempty $A \subseteq \mathfrak{C}$ for which $I_A := \{j : \text{Align}_{\Sigma}(R_j) = A\}$ is infinite. If A has a null entry, then every target projection of the corresponding source-current occurrences vanishes. This contradicts $A = \text{Align}_{\Sigma}(R_j) \neq \emptyset$. If A has a paid entry, the pairwise disjoint events $(Q_j)_{j \in I_A}$ carry the divergent lower-price schedule $(\delta_{u,A,j})_{j \in I_A}$, contradicting Theorem III.1.11. The ledger alternatives are exhaustive, so contact is impossible. $\square$

Corollary III.1.14 (Estimate-to-invariant reduction). *In the setting of Theorem III.1.13, no global estimate of the full orbit is required. It is enough, after propagating all compatible reductions, to verify for each persistent source support either the vanishing invariant (121) or the divergent physical-price schedule (122). In particular, arbitrarily large target-null motion is irrelevant to the exclusion proof.*

Corollary III.1.15 (Entropy exclusion is physical expenditure exclusion). *Suppose a canonical contact cascade carries the entropy cocycle of Definition II.9.1, its error series are finite, and $\mathfrak{D}_j(I_j) \geq \delta > 0$ on every selected window. Then contact is impossible.*

Proof. By Corollary II.9.6, the glued production is a finite physical measure. Apply Theorem III.1.11 to its restrictions to the disjoint events Q_j . $\square$

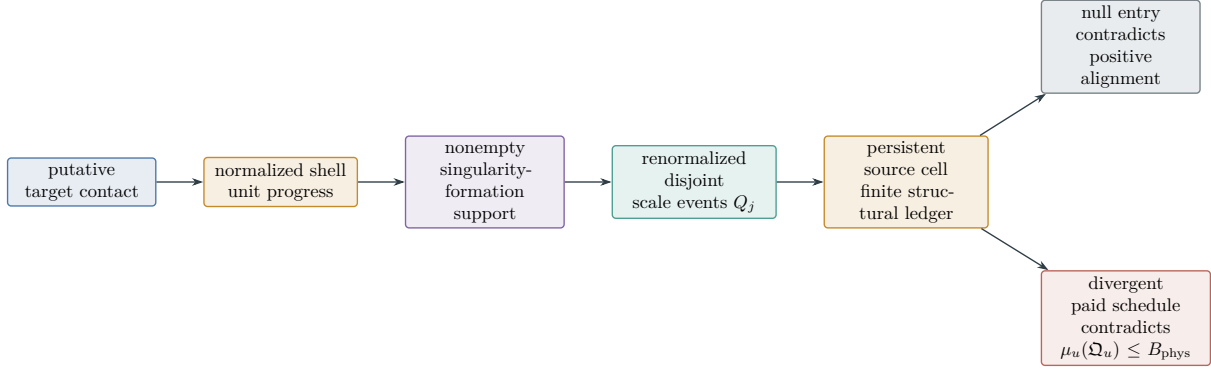


Figure 19: Singularity-formation accounting. Contact forces a unit normalized shell crossing and therefore exposes a target-aligned source. Repeated scales select a persistent source cell. A null ledger entry contradicts alignment without an estimate; a paid entry contradicts the sole finite physical budget.

III.2 The continuous structural regularity theorem

The five-channel conclusion is derived before any regularity estimate. The only equation-dependent step in target exclusion is the physical evaluation of the target-aligned class exposed by contact.

Definition III.2.1 (Physical reachability profile). For $B_{\text{phys}} < \infty$, let $\mathcal{O}_{\leq B_{\text{phys}}}$ denote all legal orbits whose expenditure measure satisfies $\mu_u(\Omega_u) \leq B_{\text{phys}}$, and assume this class is nonempty. Define

$$\text{Reach}_{\Sigma}^{\text{phys}}(B_{\text{phys}}) := \sup_{u \in \mathcal{O}_{\leq B_{\text{phys}}}} \mathbf{1}_{\{u \text{ contacts } \Sigma\}}. \quad (125)$$

Thus $\text{Reach}_{\Sigma}^{\text{phys}}(B_{\text{phys}}) = 0$ means that no legal orbit under that physical ceiling contacts the target.

Theorem III.2.2 (Unconditional continuum contact alternative). *Fix a physical continuum presentation whose native evolution satisfies one of the two interfaces of Theorem I.4.4, and close its orbit algebra sourcewise. Let Σ have a normalized shell presentation whose chain rule holds on regular pre-contact intervals. Then every target-active current and target-visible limit has ancestry in Geom , Caus , Abs , Lift , Bdry , and*

$$G_{B_{\text{phys}}, \Sigma}^{J_{\text{res}}, \text{src}, \text{sat}} = 0, \quad (126)$$

Every contact orbit has a canonical sequence of pairwise disjoint pre-contact shell intervals I_j with unit target progress. After passage to a subsequence, one nonempty support $A \subseteq \mathfrak{C}$ occurs on every I_j and contributes positive target progress there.

Proof. The continuum normal-form compiler and source-preserving completion give the five-channel statement. The target quotient and Theorem I.3.8 give (126), including for nonclosed gradient ranges. The canonical disjoint shell-cascade theorem supplies the intervals; normalized shell accountability supplies their nonempty aligned supports; and the infinite pigeonhole principle selects a persistent support. $\square$

Theorem III.2.3 (Structural physical-price exclusion). *In the setting of Theorem III.2.2, suppose every legal orbit begins outside Σ , its physical measure is defined on the canonical shell intervals,*

and $\mu_u(\Omega_u) \leq B_{\text{phys}} < \infty$. For every nonempty support that persists on a canonical contact cascade, suppose the equation's structural invariants give the implication

$$\text{Prog}_{\Sigma}^A(I_j) > 0 \implies \mu_u(I_j) \geq \delta_{u,A,j} \geq 0, \quad \sum_{j=1}^{\infty} \delta_{u,A,j} = \infty. \quad (127)$$

Then the singular target is unreachable:

$$\boxed{\text{Reach}_{\Sigma}^{\text{phys}}(B_{\text{phys}}) = 0.} \quad (128)$$

Proof. If a legal orbit contacted Σ , Theorem III.2.2 would supply pairwise disjoint intervals and a persistent support A . On their selected subsequence, (127) and countable additivity give

$$B_{\text{phys}} \geq \sum_j \mu_u(I_j) \geq \sum_j \delta_{u,A,j} = \infty,$$

a contradiction. $\square$

Remark III.2.4 (Orbitwise source ledgers). The paid branch needs a divergent lower-price schedule only for the source support that persists along a putative singular cascade. The schedule may depend on that orbit and support and may decay with scale. A uniform function $\delta_{\text{phys}}(B_{\text{phys}}) > 0$ is stronger and yields uniform window counts. Theorem III.2.3 uses only divergence of the price series on the source support selected by contact. It assumes neither five-channel exhaustion nor cascade existence; both were proved earlier.

Corollary III.2.5 (PDE global-regularity criterion). *Let every admissible initial datum $u_0 \in X_{\text{reg}}$ generate a maximal legal regular orbit on $[0, T_{\text{max}})$ with a finite physical expenditure budget. Assume the continuation criterion*

$$T_{\text{max}} < \infty \implies u \text{ contacts } \Sigma_{\text{sing}} \quad (129)$$

in the topology of Definition III.1.1. If the compiled continuum interfaces and the persistent-source price implication (127) hold for every such orbit, then $T_{\text{max}} = \infty$ for every admissible datum.

Proof. If $T_{\text{max}} < \infty$, (129) gives target contact, contradicting Theorem III.2.3. $\square$

Corollary III.2.6 (Quantitative finite-cascade bound). *If the physical expenditure satisfies $\mu_u(\Omega_u) \leq B_{\text{phys}}$ and every disjoint target-bad event costs at least $\delta_{\text{phys}}(B_{\text{phys}}) > 0$, then any such family has cardinality at most*

$$\left\lfloor \frac{B_{\text{phys}}}{\delta_{\text{phys}}(B_{\text{phys}})} \right\rfloor. \quad (130)$$

Proof. The proof of Theorem III.1.11 gives $N\delta_{\text{phys}}(B_{\text{phys}}) \leq B_{\text{phys}}$ for every finite subfamily. $\square$

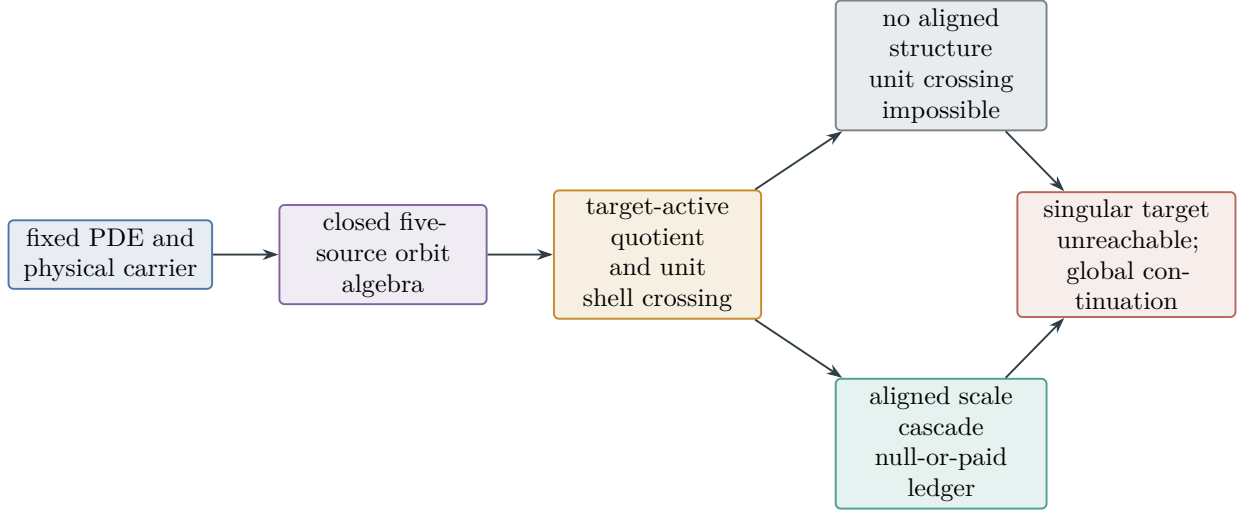


Figure 20: Structural global-regularity compiler. Only the target-active quotient is analyzed. If it contains no singularity-aligned structure, normalized shell contact is impossible directly. Otherwise, a finite null-or-paid source ledger reduces the remaining proof to structural invariants and one physical expenditure contradiction.

III.3 Certificate hierarchy and failure semantics

The following table records what each analytic module can establish. No row is allowed to substitute for another.

Module	Legitimate output	What it does not supply by itself
Target-active quotient	exact removal of every direction invisible to all prospective singularity tests	a bound on any surviving aligned direction
Propagated structural reduction	a closure space, sharp radius or gap, or target-projected equality remainder on the reduced source cell	the analytic fact that the required invariant vanishes or is positive
Closed-range realization	potential representation modulo a kernel, or an identified closure direction in the same source cell	a target-null or physical-price conclusion
Symmetric Hamiltonian/resistance	a Geom capacity or response bound	full fixed-Caus arrival or non-normal response
Committer quotient and harmonic defect	canonical binary-contact coordinate and exact resistance norm of its residual	the full hitting-time law or a nonnormal generator bound
Full resolution/pseudospectrum	a Geom + Caus response estimate	a physical event price or a singular cascade
Capacity polarity	direct target avoidance for a polar target under the declared right process	reachability from positive capacity
Commutator or twisted carrier	a physical measure controlling degenerate Geom + Caus directions	the domain, bracket-gap, or carrier inequalities
Constraint-tail normal form	local source plus gauge, summable tail, and a finite-dimensional target-projected cokernel	a physical price for the cokernel without a carrier comparison
Action-aiming projection	target-aligned part of an existing physical current	compactness or a uniform lower price
Aubin–Lions or another compactness theorem	a compact certificate topology	zero-price rigidity or target badness
Compactness plus rigidity	a uniform gap for one typed bad class and price	a globally finite physical budget
Covariant entropy cocycle	a finite physical production measure across charts	a positive price without coercivity and rigidity
Canonical shell-cascade theorem	infinitely many disjoint pre-contact events	positive expenditure on those events
Scale-cascade compiler	a physical lower-price schedule from a normalized gap and the exact scaling exponent	exclusion when the resulting scale-weight series is summable
Finite source ledger	target exclusion from null entries or divergent paid schedules on every persistent aligned source cell	the equation-specific proof of those entries

Remark III.3.1 (Incomplete certificates preserve channel type). Failure to prove compactness, target closedness, lower semicontinuity, zero-price rigidity, entropy reset compatibility, clock comparison, or a normalization or price implication means that the corresponding evaluation row has not been established. The target-visible record remains in its inherited source cell, and its saturated profile is unevaluated by that method. It does not become a nonzero J_{res} -class. This distinction is forced by Theorems I.3.4 and I.4.5 and is not an optional convention.

Remark III.3.2 (Separation of certificate roles). The logical interfaces enforce ten distinctions.

- (N1) The fixed **Caus** term is transported by a chart; it is not optimized to point toward the target.
- (N2) A killed spectral resolvent has an arrival interpretation only after a terminal-flux realization is proved.
- (N3) Finiteness of a basic energy does not imply a positive cost at every singular scale; (90) is required.
- (N4) Local entropy monotonicity does not yield a global production bound without the reset comparison (104).
- (N5) Weak compactness does not preserve a nonlinear badness predicate unless that predicate is closed in the selected topology.
- (N6) Failure of a required interface terminates that proof route with an unevaluated existing source profile, not with a new target-bearing source.
- (N7) Large or noncompact motion in $\mathcal{M}_\Sigma$ requires no downstream estimate because it cannot increase a declared prospective target functional.
- (N8) A quotient is propagated only when every downstream map factors through it; gauge fixing alone does not justify discarding a direction.
- (N9) Summability of a constraint tail closes the analytic tail but supplies a physical price only through a carrier comparison such as (102).
- (N10) A positive normalized scale gap excludes an infinite cascade when the pulled-back lower prices have the divergent total mass required by (117); a nonvanishing infimum is a stronger sufficient condition.

III.4 A reusable PDE instantiation protocol

For a particular deterministic PDE, the abstract theory is instantiated in the following order.

- (I1) Specify the regular orbit class, maximal interval, singular germ target, and continuation criterion.
- (I2) Identify a lower-bounded physical carrier and derive its exact balance. Energy, action, localized dissipation, commutator production, boundary work, and entropy production may contribute, but the carrier compiler must produce the single nonnegative measure μ_u .
- (I3) Label the native mobility or reversible form **Geom**, the fixed directed transport **Caus**, quotient operations **Abs**, blow-up charts and augmented variables **Lift**, and terminal or exterior fluxes **Bdry**.
- (I4) Form the sourcewise mathematical closure of every dynamically generated weak limit, concentration, oscillation, tail, defect, and blow-up germ.
- (I5) Construct normalized target shells and verify the pre-contact chain rule. Contact then supplies unit shell progress by Theorem I.5.2. Killed-generator and predictable-compensator identities may supply additional source factorizations. Compute the resulting singularity-formation support $\mathfrak{S}_\Sigma$.
- (I6) Propagate every certified target-null, gauge, conservation, and equality direction through all compatible later quotients. If the formation support is empty, apply Proposition III.1.3 and

stop.

- (I7) Resolve any nonlocal constraint into its local source, gauge, summable exterior tail, and finite target-projected cokernel. Resolve degenerate Geom directions by a closed-range, commutator, or capacity invariant where applicable.
- (I8) On the surviving aligned branch, prove that contact generates infinitely many pairwise disjoint normalized bad events and record the exact physical scaling exponent.
- (I9) Complete the finite source ledger. For each persistent aligned source cell, prove either a target-null closure invariant or a lower physical-price schedule with divergent sum, using a sharp radius, resistance, compactness–rigidity gap, equality separator, scale law, entropy cocycle, or a direct carrier inequality.
- (I10) Apply Theorem III.2.3 and the continuation criterion.

This protocol is equation-agnostic. Navier–Stokes, geometric flows, reaction–diffusion systems, kinetic equations, and dispersive–dissipative systems may require different choices in steps (I2), (I5), (I7), (I8), and (I9). The closed source algebra and failure semantics do not change.

III.5 Conclusion

The continuum completion of structural complexity is a theory of physical reachability. Its closure contains every state, profile, measure, trace, and germ generated by legal finite-budget orbits. Limit formation is source-preserving, and the residual is defined only after saturation as a target-null kernel. Hence a target-visible continuum residual cannot exist.

Normalized shell accountability writes unit target progress as an exact integral of the pre-contact current and proves that a singularity requires nonempty target-aligned source support. When that support is empty, global target avoidance follows without estimating the size of the target-null motion. When it is nonempty, propagated reductions remove every already closed direction, and Hamiltonian, resolvent, carrier, capacity, constraint-tail, compactness, scale, and entropy arguments reduce to a finite set of closure, radius, gap, and equality invariants.

The structural regularity theorem therefore asks less than a global estimate of every solution component. A contact cascade selects a persistent source cell; a null ledger entry contradicts its alignment, while a paid entry forces the lower physical prices of infinitely many disjoint events to have divergent total mass in one globally finite measure. When a required invariant has not been proved, regularity is open because an existing aligned profile has not been evaluated, not because the continuum has generated a sixth source.

A Dependency table for the main conclusions

Conclusion	Required hypotheses	Not required
Source-preserving completion	sourcewise closure in a Hausdorff test system	compactness of the whole orbit class
Markov source compilation	regular symmetric Dirichlet form, Beurling–Deny data, and sector-bounded antisymmetric form	a global regularity or event-price estimate

Zero target-visible residual	continuous target tests and residual defined as their common kernel	a positive price or PDE regularity
Target-alignment necessity	normalized target shells, the pre-contact chain rule, and the compiled current normal form	compactness or a physical budget
Direct target avoidance	normalized shell crossing and empty singularity-formation support	a cascade or event-price estimate
Deterministic accountability	absolute continuity of shell observables and the stated current split	Hamiltonian or Markov structure
Resolvent accountability	common resolvent point, domain-compatible generator split, and target-null residual pairing	self-adjointness of the full generator
Stochastic accountability	Doob–Meyer compensator and integrable source split	a density or diffusion representation
Carrier physical budget	lower-bounded carrier, measure balance, passivity, and finite positive input	coercivity of the full state norm
Structural reduction	closed compatible quotient and descended downstream maps	an estimate of discarded target-null directions
Uniform physical price	one typed compactness–rigidity certificate and physical domination	an entropy primitive
Constraint-tail closure	local elliptic split, summable decaying tail, closed gauge range, and finite low space	a physical price without a carrier comparison
Scale-cascade price	disjoint scale extraction, normalized gap, physical pull-back, and divergent pulled-back weight series	a global bound for every orbit component
Finite entropy production	local production, lower scalar bound, and summable reset/error terms	a global adjoint density
Target exclusion	exact shell accountability and either empty aligned support or a persistent aligned support with a structurally proved divergent physical price schedule	a closed formula or global norm estimate for every limit profile
PDE global regularity	target exclusion plus the declared continuation criterion	a universal PDE-independent compactness theorem

B Notation index

Symbol	Meaning
$\mathfrak{C}$	the five-source alphabet $\{\text{Geom}, \text{Caus}, \text{Abs}, \text{Lift}, \text{Bdry}\}$
$\widehat{X}_\Sigma, U_r(\Sigma), \Phi_r$	target state-or-germ space, normalized shell neighborhoods, and shell observables
B_{phys}	sole physical expenditure ceiling for a legal orbit
$\mathcal{R}_A^{\text{phys}, \text{sat}}(B_{\text{phys}})$	sourcewise saturated physical record cell with support A
$\text{Anc}(R), \text{Src}(R)$	full ancestry word and its elementary source support
$\mathcal{K}_\Sigma$	common kernel of all prospective target functionals
J_{res}	target-null post-saturation residual coordinate; never a source
$\text{Align}_\Sigma(R)$	source labels with positive target progress in a prospective record
$Q_{c,r}(u), \text{Prog}_\Sigma^c(R)$	signed contribution of source c to a normalized shell crossing or another prospective factorization
$\mathfrak{S}_\Sigma(u)$	union of aligned labels in the orbit's shell-complete family; its singularity-formation support
μ_u	finite nonnegative localizable physical expenditure measure for one orbit
α_*, δ_*	sharp domination radius and normalized physical price gap on a reduced source space
$\mathfrak{H}_\mathcal{F}^\Sigma$	closed target projection of a candidate source family
β_*^{red}	reduced passivity radius of feeding flux relative to gross expenditure
$\mathfrak{o}_{\text{tail}}$	finite-dimensional gauge-cokernel class of an elliptic constraint tail
κ	physical pullback exponent for a normalized scale event
B_Σ	closed normalized target-approaching class in a certificate topology
$\mathfrak{d}$	lower-semicontinuous physical price functional on normalized windows
δ_{phys}	positive lower physical price for a selected event
$\delta_{u,A,j}$	scale-dependent lower-price schedule for a persistent aligned source row; its paid-ledger sum diverges
$\text{Reach}_\Sigma^{\text{phys}}(B_{\text{phys}})$	physical target-contact profile under the single ceiling B_{phys}

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